

From Local Response to Global Momentum: Asymptotic Boundary Completion in Motion–Staticity Geometry

Mingyang Han

School of Mathematical Sciences, Shanghai Jiao Tong University

Shanghai, 200240, China

hanmingyang@sjtu.edu.cn

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Abstract The motion–staticity (MS) programme reconstructs relativistic massive-generator geometry from a compact observer state and, through a scalar generator–displacement response, reconstructs a nondegenerate solder map and Lorentzian spacetime geometry. In the restricted classical branch developed previously, this local construction also supplies the bulk hypersurface-deformation generators. It does not, however, define the total energy–momentum of an isolated gravitating system.

We study the corresponding local-to-global problem on calibrated asymptotically Minkowskian ends. Standard Regge–Teitelboim falloff and parity conditions are imposed, while the operational response is additionally calibrated at the asymptotic orders relevant to translation charges. Allowing the reconstructed bulk generators to approach constant translations produces a surface obstruction to functional differentiability. Its boundary primitive defines a total translation covector

$$P_{\infty}^{\text{MS}} \in t_{\infty}^*.$$

The reduced translation charges form an Abelian Hamiltonian sector and are conserved under admissible translation evolution.

The resulting global charge admits an observer-space representation closely parallel to the local massive-response polarity. For the associated positive-energy vector P_{∞}^{MS} , the scalar observer energy

$$\mathcal{E}_{\infty}(u) = -\eta_{\infty}(P_{\infty}^{\text{MS}}, u)$$

determines the complete vector through

$$P_{\infty}^{\text{MS}} = \mathcal{E}_{\infty} u - \nabla_H \mathcal{E}_{\infty},$$

while

$$-\eta_{\infty}(P_{\infty}^{\text{MS}}, P_{\infty}^{\text{MS}}) = \mathcal{E}_{\infty}^2 - |\nabla_H \mathcal{E}_{\infty}|^2.$$

In the future-timelike sector, the invariant radial quantity is the unique minimum of the observer-energy function. This gives a local–global response reciprocity: locally one varies the massive generator at fixed displacement to obtain the proper-time norm, whereas globally one varies the observer at fixed total generator to obtain the invariant mass.

We also identify the translation-sensitive modes of the asymptotic response calibration. The monopole of the leading r^{-1} scale mode shifts the energy, while the dipole of the r^{-2} normal first jet shifts the spatial translation charge. Finally, after imposing the empirical Einstein normalization

$$\kappa = \frac{8\pi G}{c^4},$$

the completed MS translation charge agrees with the ADM four-momentum and its radial invariant agrees with the ADM mass. The ADM boundary mechanism, the hyperbolic observer identities, and the positive-mass theorem are not claimed as new results; the contribution is their operational provenance and structural relation within the MS construction.

Keywords: motion–staticity geometry; asymptotic flatness; Hamiltonian boundary terms; ADM energy–momentum; Lorentzian response geometry; invariant mass

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1 Introduction

A local generally covariant theory does not by itself supply the total energy or momentum of an isolated gravitating system. In canonical gravity the bulk Hamiltonian and momentum constraints generate hypersurface deformations, but nontrivial translation charges arise only after an asymptotic structure has been specified and the generators have been made functionally differentiable. For asymptotically flat general relativity this is the classical Regge–Teitelboim mechanism, whose translation-sector surface terms give the ADM energy–momentum [7, 1, 2].

The purpose of the present paper is not to construct a different ADM surface functional. It is to ask where the asymptotic translation charge appears when the bulk gravitational variables and generators have already arisen from the preceding motion–staticity (MS) construction.

The starting point of that programme is the compact local state

$$(D, S) \in \mathbb{R}^3 \oplus \mathbb{R}, \quad |D|^2 + S^2 = 1, \quad S > 0, \quad (1)$$

with projective image

$$\Phi(D, S) = \frac{1}{S}(1, D). \quad (2)$$

In Ref. [3], the operational interpretation of this state yields the flat Lorentzian chronometric structure, while the dynamical rigidity analysis selects the standard relativistic free-particle Hamiltonian and positive massive generator shell. Under the additional observer-completeness assumptions stated there, the connected Lorentz observer structure is recovered. The one-particle generator is naturally dual to the flat translation algebra rather than introduced only through a postulated four-vector transformation law.

The second paper moves from this flat translation geometry to a curved spacetime setting [5]. At each event x , the internal Lorentz generator space is kept distinct from the spacetime tangent space, and the missing relation between them is reconstructed from the scalar generator–displacement response

$$\widetilde{\Theta}_x(\Pi, \xi) = \Pi(e_x \xi). \quad (3)$$

Under the operational assumptions specified there, this response determines a unique nondegenerate solder map

$$e_x : T_x M \xrightarrow{\sim} V_x \quad (4)$$

and hence

$$g = e^* \eta. \quad (5)$$

Within the restricted torsion-free metric branch considered in the same work, the reconstructed geometry admits the standard hypersurface-deformation form of the gravitational bulk generator. This is a restricted classical completion, not a derivation of all possible gravitational dynamics.

The third companion paper follows a different branch [4]. There the same soldering response is promoted to a family of configuration observables and, after an additional coherent deformation principle, to a canonical quantum algebra. The present work does not use that quantum completion.

It returns instead to the classical branch and asks how a locally reconstructed gravitational geometry acquires a global translation generator when the system is asymptotically isolated.

This question is not answered by the existence of the local massive shell. In a generic curved spacetime there is no global Abelian translation group, and generator covectors belonging to distinct spacetime points cannot be canonically added by parallel transport without further structure. Moreover, the total gravitational mass is not a sum of local one-particle rest scales. The appropriate global object must instead be associated with the Hamiltonian realization of the asymptotic remnant of translation symmetry.

We therefore consider a spacelike hypersurface with an asymptotically Minkowskian end and choose an absolutely calibrated flat reference response. The reconstructed canonical data are required to satisfy the standard Regge–Teitelboim falloff and parity conditions. Constant leading lapse and shift define the quotient translation space

$$\mathfrak{t}_\infty \simeq \mathbb{R} \oplus \mathbb{R}^3. \quad (6)$$

At this stage $\mathfrak{t}_\infty$ contains translation parameters only; no total momentum has yet been defined.

The first problem of the paper is therefore Hamiltonian: given $a \in \mathfrak{t}_\infty$, can the reconstructed bulk generator be extended from the compact-support sector to a finite and differentiable generator of that asymptotic translation? The answer requires a boundary completion

$$G[a] = G_0[a] + B_\infty[a]. \quad (7)$$

Because $B_\infty[a]$ is linear in the asymptotic translation parameter, it defines

$$\mathbf{P}_\infty^{\text{MS}} \in \mathfrak{t}_\infty^*, \quad \mathbf{P}_\infty^{\text{MS}}(a) = B_\infty[a]. \quad (8)$$

This is the point at which a global momentum-like object first enters the construction.

The boundary calculation uses standard asymptotically flat Hamiltonian methods. Its role here is nevertheless different from taking the ADM four-momentum as primitive data. The generator being completed occurs only after the sequence

$$\text{compact massive state} \longrightarrow \text{operational response} \longrightarrow \text{soldered geometry} \longrightarrow \text{restricted bulk generator}. \quad (9)$$

The boundary analysis continues this sequence to

$$\text{restricted bulk generator} \longrightarrow \text{asymptotic boundary completion} \longrightarrow \mathbf{P}_\infty^{\text{MS}}. \quad (10)$$

This operational provenance is the first main structural point of the paper.

The second concerns the relation between local and global Lorentzian response geometry. The local massive response determines, for a future timelike displacement,

$$\mu_x(\xi) = \frac{1}{Q} \inf_{\widehat{\Pi} \in \widehat{\Sigma}_{Q,x}^+} \widehat{\Pi}(e_x \xi) = \sqrt{-g_x(\xi, \xi)}. \quad (11)$$

The underlying Lorentzian minimization is standard. Its importance here is that the shell over which the minimum is taken is precisely the massive generator sector inherited from the preceding MS construction.

After the global boundary charge has been obtained, the same Lorentzian polarity reappears with the roles exchanged. Let $\mathbf{P}_\infty^{\text{MS}}$ be the positive-energy vector Lorentz-polar to the completed translation

covector, and let $\mathcal{H}_\infty^+$ denote the asymptotic future unit observer hyperboloid. The scalar energy seen by observer u is

$$\mathcal{E}_\infty(u) = -\eta_\infty(P_\infty^{\text{MS}}, u). \quad (12)$$

For a future-timelike total charge,

$$\inf_{u \in \mathcal{H}_\infty^+} \mathcal{E}_\infty(u) = \sqrt{-\eta_\infty(P_\infty^{\text{MS}}, P_\infty^{\text{MS}})}. \quad (13)$$

Thus the local and global constructions exchange which object is varied:

$$\begin{aligned} \text{local:} & \quad \text{fixed displacement} + \text{variable massive generator,} \\ \text{global:} & \quad \text{fixed total generator} + \text{variable asymptotic observer.} \end{aligned} \quad (14)$$

We refer to this structural relation as *local–global response reciprocity*. It is not a new Lorentzian inequality; it is a relation between two stages of the same MS construction.

The observer-space formulation also makes the information content of the total charge particularly transparent. On the unit hyperboloid,

$$P_\infty^{\text{MS}} = \mathcal{E}_\infty u - \nabla_H \mathcal{E}_\infty, \quad (15)$$

and

$$-\eta_\infty(P_\infty^{\text{MS}}, P_\infty^{\text{MS}}) = \mathcal{E}_\infty^2 - |\nabla_H \mathcal{E}_\infty|_{h_H}^2. \quad (16)$$

These are standard hyperbolic identities for ambient Lorentzian linear functionals. In the present setting they show that a scalar observer-energy field provides a complete readout of a translation vector that was produced independently by Hamiltonian boundary completion.

The third main point concerns the asymptotic calibration of the operational response. A positive rescaling

$$\Theta \mapsto \lambda \Theta \quad (17)$$

preserves the causal cone but changes the reconstructed metric scale. At the asymptotic orders relevant to translations, we find that

$$\lambda - 1 = \frac{\sigma(\hat{x})}{r} + \dots \quad (18)$$

changes the total energy only through the monopole of σ , whereas

$$n(\ln \lambda) = \frac{\rho(\hat{x})}{r^2} + \dots \quad (19)$$

changes the spatial translation charge only through the dipole of ρ . The familiar scalar-plus-vector representation of translations is standard. The distinctive statement here is that these four charge-supporting modes arise as different components of the asymptotic operational response jet.

Response calibration must also be distinguished from gravitational normalization. Once the asymptotic geometry is held fixed, the boundary charge scales as

$$P_\infty^{\text{MS}} \propto \kappa^{-1}, \quad (20)$$

whereas its projective rest-observer direction is unchanged under a common positive rescaling of κ . The construction therefore separates the projective direction of the total momentum from its radial

gravitational magnitude. This parallels, without identifying, the separation between the compact one-particle state and its shell scale in the flat theory.

Only after these structures have been obtained do we impose the empirical Einstein normalization

$$\kappa = \frac{8\pi G}{c^4}. \quad (21)$$

The resulting energy and physical spatial momentum then agree with the ADM quantities,

$$E_\infty^{\text{MS}} = E_{\text{ADM}}, \quad \mathbf{P}_\infty^{\text{MS}} = \mathbf{P}_{\text{ADM}}, \quad (22)$$

and the future-timelike radial invariant gives

$$M_\infty^{\text{MS}} = M_{\text{ADM}}. \quad (23)$$

The value of G is not derived. Nor is the local one-particle scale Q/c identified with the total gravitational mass.

The scope of the paper is correspondingly limited. We construct only the four-dimensional asymptotic translation sector. Rotations, boosts, angular momentum, center-of-mass charges, and enlarged BMS-type sectors require additional boundary analysis. We do not derive a positive-mass theorem from the MS compact-state relation. The complete Lorentzian causal classification of the translation vector is developed independently of positivity; after the ADM identification, standard positive-energy and rigidity theorems may be invoked under their own hypotheses. Finally, the quantum soldering branch of Ref. [4] is not used to quantize the boundary charge in the present work.

The three contributions emphasized in the paper can therefore be stated concisely:

$$\begin{aligned} \text{local operational geometry} &\longrightarrow \text{boundary-completed total translation covector,} \\ \text{local massive-response polarity} &\longleftrightarrow \text{global observer-response polarity,} \\ \text{asymptotic response jet} &\longrightarrow \text{translation-sensitive charge modes.} \end{aligned} \quad (24)$$

The remainder of the paper is organized as follows. Section 2 isolates the local operational Lorentzian polarity and fixes the vector–covector conventions inherited from the companion papers. Section 3 defines the calibrated asymptotically Minkowskian phase space and the asymptotic translation space. Section 4 constructs the differentiable translation generator and the total translation covector. Section 5 establishes the reduced Abelian translation sector, Lorentz covariance, and conservation. Section 6 develops the observer-space representation and the local–global response reciprocity. Section 7 identifies the charge-supporting response modes, restores physical units, and establishes the Einstein-normalized ADM realization. Section 8 summarizes the scope and open extensions.

2 Operational Response Polarity

The purpose of this section is to isolate the local Lorentzian structure that will later be used to interpret the asymptotic translation charge. The logical input comes from the first two companion papers, while the present paper follows the classical branch of the construction.

Ref. [3] selects the standard relativistic massive generator geometry from the compact motion–staticity state. In the signature convention used there, the corresponding massive shell is represented with the opposite overall Lorentzian sign. In the present paper we instead follow Refs. [5, 4] and use

$$\text{sign}(\eta_x) = (-, +, +, +). \quad (25)$$

Whenever physical units are displayed, the positive-energy momentum vector and its canonical action covector are therefore

$$P^a = \left(\frac{E}{c}, \mathbf{p} \right), \quad \Pi_a = P_a^b = \left(-\frac{E}{c}, \mathbf{p} \right). \quad (26)$$

The distinction between P and Π will be maintained throughout. Except where needed for comparison with the companion papers, we set $c = 1$ in the local formulas below.

The second companion paper reconstructs, at every event $x \in M$, a nondegenerate solder map

$$e_x : T_x M \xrightarrow{\sim} V_x \quad (27)$$

from the operational generator–displacement response

$$\widetilde{\Theta}_x(\Pi, \xi) = \Pi(e_x \xi), \quad \Pi \in V_x^*, \quad \xi \in T_x M, \quad (28)$$

and hence determines

$$g_x = e_x^* \eta_x. \quad (29)$$

These statements are established results of Ref. [5]; they are not rederived here. The third companion paper takes the same scalar family Eq. (28) and regards it as a family of configuration observables while e varies in soldering space [4]. The present work follows the other branch: e and g are treated classically, and the question is how the corresponding bulk gravitational generator acquires a global translation charge on an asymptotically isolated spacetime.

Before addressing that boundary problem, it is useful to express the Lorentzian geometry of Eq. (29) directly in terms of the same operational response. This will also make explicit which part of the construction is projective and which part requires an absolute response calibration.

2.1 Local massive-response polarity

Let V_x be the four-dimensional internal Lorentz space at x , and let

$$\mathcal{H}_{Q,x}^+ := \left\{ P \in V_x : \eta_x(P, P) = -Q^2, \quad P \text{ future directed} \right\}, \quad Q > 0, \quad (30)$$

denote the positive massive vector shell inherited from the flat relativistic sector. For $P \in \mathcal{H}_{Q,x}^+$, define

$$\Pi := P^b, \quad \widehat{\Pi} := -\Pi = -P^b. \quad (31)$$

The covector Π is the canonical action covector used in Refs. [5, 4]. The sign-reversed covector $\widehat{\Pi}$ is convenient for positive response on future-directed timelike displacements. We denote its shell by

$$\widehat{\Sigma}_{Q,x}^+ := \left\{ -P^b : P \in \mathcal{H}_{Q,x}^+ \right\} \subset V_x^*. \quad (32)$$

Thus

$$\eta_x^{-1}(\widehat{\Pi}, \widehat{\Pi}) = -Q^2. \quad (33)$$

The sign change in Eq. (31) does not introduce a second physical momentum variable. It separates the canonical action convention from the positive-response convention required for the variational construction below. For $\widehat{\Pi} = -\Pi$, define

$$\Theta_x^+(\widehat{\Pi}, \xi) := -\widetilde{\Theta}_x(\Pi, \xi) = \widehat{\Pi}(e_x \xi). \quad (34)$$

Let

$$T_x^+ := \{\xi \in T_x M : g_x(\xi, \xi) < 0, \quad \xi \text{ future directed}\} \quad (35)$$

be the open future-timelike cone. For $\xi \in T_x^+$, introduce the normalized massive-response polar functional

$$\mu_x(\xi) := \frac{1}{Q} \inf_{\widehat{\Pi} \in \widehat{\Sigma}_{Q,x}^+} \Theta_x^+(\widehat{\Pi}, \xi). \quad (36)$$

Equivalently, after transporting the positive-response shell to the physical cotangent space,

$$\widehat{\Sigma}_{Q,x}^{\text{phys},+} := e_x^* \widehat{\Sigma}_{Q,x}^+ \subset T_x^* M, \quad (37)$$

one has

$$\mu_x(\xi) = \frac{1}{Q} \inf_{\widehat{p} \in \widehat{\Sigma}_{Q,x}^{\text{phys},+}} \widehat{p}(\xi). \quad (38)$$

The Lorentzian minimization underlying the following proposition is standard. The point specific to the present construction is that the dual shell entering the minimization is not introduced independently: it is the massive generator sector already selected in Ref. [3] and coupled operationally to spacetime displacements in Ref. [5].

Proposition 2.1 (Operational realization of Lorentzian polarity). *For every $\xi \in T_x^+$,*

$$\mu_x(\xi) = \sqrt{-g_x(\xi, \xi)}. \quad (39)$$

The infimum is attained by a unique $\widehat{\Pi}_\xi \in \widehat{\Sigma}_{Q,x}^+$. The corresponding physical covector is

$$\widehat{p}_\xi := e_x^* \widehat{\Pi}_\xi = -\frac{Q}{\mu_x(\xi)} g_x^b(\xi). \quad (40)$$

Proof. Set

$$Y := e_x \xi.$$

Since e_x transports the chosen time orientation, Y is future timelike. Every $\widehat{\Pi} \in \widehat{\Sigma}_{Q,x}^+$ has the form

$$\widehat{\Pi} = -P^b$$

for a unique $P \in \mathcal{H}_{Q,x}^+$. Writing

$$U := \frac{P}{Q}, \quad \eta_x(U, U) = -1,$$

Eq. (34) gives

$$\frac{1}{Q} \Theta_x^+(\widehat{\Pi}, \xi) = -\eta_x(U, Y). \quad (41)$$

The reverse Lorentzian Cauchy–Schwarz inequality for two future-directed timelike vectors yields

$$-\eta_x(U, Y) \geq \sqrt{-\eta_x(Y, Y)}, \quad (42)$$

with equality precisely when

$$U = \frac{Y}{\sqrt{-\eta_x(Y, Y)}}.$$

Using $g_x = e_x^* \eta_x$ gives Eq. (39). The equality condition fixes P , hence $\widehat{\Pi}$, uniquely. Pulling $-P^b$ back with e_x^* gives Eq. (40). $\square$

The minimizing covector is also recovered from the first derivative of the same scalar response functional. Differentiating Eq. (39) with respect to the vector-space argument gives

$$d\mu_x|_{\xi}(v) = -\frac{g_x(\xi, v)}{\mu_x(\xi)}, \quad v \in T_x M, \quad (43)$$

and therefore

$$\widehat{p}_{\xi} = Q d\mu_x|_{\xi}. \quad (44)$$

Thus the response minimum defines a Gauss-type map

$$T_x^+ \longrightarrow \widehat{\Sigma}_{Q,x}^{\text{phys},+}, \quad \xi \longmapsto \widehat{p}_{\xi}. \quad (45)$$

Geometrically, $\widehat{p}_{\xi}$ is normal to the level hypersurface of μ_x through ξ inside the future-timelike cone of $T_x M$. No spacetime foliation is assumed in this statement.

The second derivative of the same scalar functional recovers the complete Lorentzian bilinear form. Since

$$\mu_x^2(\xi) = -g_x(\xi, \xi),$$

one obtains

$$g_x(v, w) = -\frac{1}{2} D^2(\mu_x^2)_{\xi}(v, w), \quad \xi \in T_x^+, \quad (46)$$

where D^2 denotes the ordinary Hessian with respect to the linear variable in $T_x M$. Equivalently,

$$g_x = -\mu_x D^2 \mu_x - d\mu_x \otimes d\mu_x. \quad (47)$$

Although the right-hand sides are evaluated at an arbitrary timelike ξ , they reconstruct the same bilinear form g_x .

The local information may therefore be organized as

$$\Theta_x \longrightarrow \mu_x \longrightarrow (d\mu_x, D^2 \mu_x) \longrightarrow (\widehat{p}_{\xi}, g_x). \quad (48)$$

The tensorial metric is not replaced by the scalar response; rather, its Lorentzian geometry is recoverable from the value and derivatives of that response polar.

2.2 Causal spectrum, projective structure, and calibration

The same infimum can be considered for arbitrary $\xi \in T_x M$, without assuming timelikeness. Define

$$\mathfrak{m}_x(\xi) := \inf_{\widehat{\Pi} \in \widehat{\Sigma}_{Q,x}^+} \frac{\Theta_x^+(\widehat{\Pi}, \xi)}{Q}. \quad (49)$$

Elementary Lorentzian geometry gives

$$\mathfrak{m}_x(\xi) = \begin{cases} \sqrt{-g_x(\xi, \xi)}, & \xi \text{ future timelike,} \\ 0, & \xi \text{ nonzero future null,} \\ 0, & \xi = 0, \\ -\infty, & \xi \text{ spacelike or nonzero past directed.} \end{cases} \quad (50)$$

For a nonzero future-null vector the value zero is an infimum, not a minimum. The minimizing sequence requires increasingly boosted massive generators and therefore approaches the projective boundary of the massive hyperboloid. In the compact MS variables of Ref. [3], this is precisely

$$S \longrightarrow 0^+. \quad (51)$$

Thus the same compact boundary that represents infinite rapidity in the flat massive theory reappears here as the ideal limiting sector of the local response-polar problem. No finite massive generator is identified with a null generator.

To distinguish the open timelike cone from its causal closure, define

$$C_x^+ := \overline{T_x^+}. \quad (52)$$

Equation (50) gives the basis-independent characterization

$$C_x^+ = \{\xi \in T_x M : m_x(\xi) \geq 0\} = \{\xi \in T_x M : m_x(\xi) > -\infty\}. \quad (53)$$

Its interior is T_x^+ ; its nonzero boundary consists of the future-null vectors; and its apex is the zero vector.

On T_x^+ , the polar functional is positively homogeneous,

$$\mu_x(\alpha\xi) = \alpha\mu_x(\xi), \quad \alpha > 0, \quad (54)$$

and satisfies the reverse triangle inequality

$$\mu_x(\xi + \zeta) \geq \mu_x(\xi) + \mu_x(\zeta), \quad \xi, \zeta \in T_x^+, \quad (55)$$

with equality precisely for positively collinear timelike vectors. Equivalently,

$$\mu_x(t\xi + (1-t)\zeta) \geq t\mu_x(\xi) + (1-t)\mu_x(\zeta), \quad 0 \leq t \leq 1. \quad (56)$$

These are standard properties of the Lorentzian proper-time functional. Their role here is to show that the operational massive response reproduces the full local causal polarity without requiring a preferred internal Lorentz frame.

Indeed, under an internal Lorentz gauge transformation

$$e_x \longmapsto e'_x = \Lambda_x e_x, \quad P \longmapsto P' = \Lambda_x P, \quad \Lambda_x \in SO^+(1, 3), \quad (57)$$

the action and positive-response covectors transform dually,

$$\Pi \longmapsto (\Lambda_x^{-1})^* \Pi, \quad \widehat{\Pi} \longmapsto (\Lambda_x^{-1})^* \widehat{\Pi}, \quad (58)$$

and the scalar response is unchanged. Consequently,

$$\mu'_x = \mu_x, \quad g'_x = g_x. \quad (59)$$

A positive rescaling of the operational response has a different status. Let

$$\widetilde{\Theta}_x \longmapsto \widetilde{\Theta}'_x = \lambda(x) \widetilde{\Theta}_x, \quad \lambda(x) > 0. \quad (60)$$

The soldering reconstruction then gives

$$e_x \mapsto e'_x = \lambda(x)e_x, \quad g_x \mapsto g'_x = \lambda(x)^2 g_x, \quad (61)$$

and therefore

$$\mu_x \mapsto \mu'_x = \lambda(x)\mu_x. \quad (62)$$

The causal cone C_x^+ is unchanged, while the proper-time and metric normalization are changed. Thus

$$\begin{aligned} \text{internal Lorentz gauge :} & \quad g_x \text{ and } \mu_x \text{ unchanged,} \\ \text{positive response recalibration :} & \quad g_x \mapsto \lambda^2 g_x, \quad \mu_x \mapsto \lambda \mu_x. \end{aligned} \quad (63)$$

The distinction will be essential at spatial infinity. The projective response geometry determines causal directions without an absolute scale, whereas a dimensionful total translation charge requires an asymptotic calibration at precisely the orders that support the boundary Hamiltonian.

There is also a useful local–global structural preview. In the present section a fixed spacetime displacement ξ is probed by varying the positive massive generator:

$$\text{fixed } \xi \quad + \quad \text{variable } \widehat{\Pi} \quad \longrightarrow \quad \mu_x(\xi). \quad (64)$$

After the asymptotic boundary completion has produced a fixed total translation generator, the roles will be reversed: the generator will be held fixed while the asymptotic observer is varied. The resulting observer-energy minimum will select its Lorentzian radial invariant. The two variational problems use the same Lorentzian polarity but occupy opposite ends of the construction. This local–global reciprocity will be made precise only after the total boundary charge has been constructed.

The output of the present section is therefore local and deliberately scale-sensitive:

$$\text{massive operational response} \longrightarrow \mu_x \longrightarrow (C_x^+, g_x), \quad (65)$$

together with a sharp distinction between internal Lorentz gauge and absolute response calibration. The next section uses this distinction to define the calibrated asymptotically Minkowskian sector on which global translation charges can be meaningfully constructed.

3 Calibrated Asymptotic Response Geometry

The local construction of Section 2 determines the causal cone and Lorentzian metric from the operational massive response without selecting an internal Lorentz frame. A total energy–momentum problem requires additional structure. In particular, an isolated configuration must be compared with an asymptotic vacuum, the absolute normalization of the response must be fixed at the orders relevant to the boundary generator, and one must distinguish deformations that survive as physical translations from those that remain proper gauge.

These requirements are not consequences of the compact MS state alone. They specify the asymptotically isolated sector in which the local response–soldering construction can be extended to a global Hamiltonian problem. We adopt the standard Regge–Teitelboim asymptotic class [7, 2], supplemented by an MS-specific calibration condition on the asymptotic response. The Regge–Teitelboim falloff and parity conditions are standard and are not claimed as new results.

Throughout this section and the boundary calculation of the next section we set $c = 1$. Physical units will be restored only after the canonical translation charge has been constructed.

3.1 Calibrated asymptotically Minkowskian data

Let Σ be a spacelike hypersurface with one asymptotic end

$$\Sigma_\infty \simeq \mathbb{R}^3 \setminus B_R, \quad (66)$$

and choose asymptotically Cartesian coordinates x^i on this end. Write

$$r := \sqrt{\delta_{ij}x^i x^j}, \quad \hat{x}^i := \frac{x^i}{r}. \quad (67)$$

The calibrated reference spatial metric and its Levi–Civita derivative will be denoted by

$$\bar{h}_{ij} = \delta_{ij}, \quad \bar{D}_i. \quad (68)$$

For a tensor field T , the notation

$$T = O_k(r^{-p}) \quad (69)$$

means that

$$\bar{D}_{i_1} \cdots \bar{D}_{i_m} T = O(r^{-p-m}), \quad 0 \leq m \leq k. \quad (70)$$

The reference geometry is tied to the operational response rather than introduced as an unrelated tetrad background. Choose an absolutely calibrated flat response

$$\bar{\Theta} \quad (71)$$

whose polar functional reconstructs the Minkowski metric

$$\bar{g} = -\frac{1}{2}D^2(\bar{\mu}^2). \quad (72)$$

The hypersurface is required to approach a standard spacelike hyperplane of this calibrated reference spacetime,

$$\bar{h}_{ij} = \delta_{ij}, \quad \bar{K}_{ij} = 0. \quad (73)$$

For the physical response Θ , let

$$g[\Theta] = -\frac{1}{2}D^2(\mu[\Theta]^2) \quad (74)$$

denote the reconstructed spacetime metric, and let h_{ij} be its pullback to Σ . Define

$$q_{ij} := h_{ij} - \bar{h}_{ij} = h_{ij} - \delta_{ij}. \quad (75)$$

The canonical momentum used below is not introduced independently of the preceding MS construction. In the minimal torsion-free metric branch of Ref. [5], the hypersurface-normal boost sector of the Lorentz connection determines the extrinsic curvature. With the convention

$$K_{ij} = \frac{1}{2N}(\dot{h}_{ij} - \mathcal{L}_\beta h_{ij}), \quad (76)$$

the restricted canonical completion gives

$$\pi^{ij} = \frac{\sqrt{h}}{2\kappa}(K^{ij} - Kh^{ij}), \quad K := h^{ij}K_{ij}. \quad (77)$$

Here π^{ij} is a symmetric contravariant tensor density of weight +1. The sign convention in Eq. (76) will be kept fixed throughout the paper; it determines the sign relating the canonical spatial boundary coefficient to the conventional physical ADM momentum.

Because the reference end is Minkowskian, the present asymptotic sector belongs to the

$$\Lambda = 0 \quad (78)$$

branch of the restricted gravitational completion. No asymptotically de Sitter or anti-de Sitter sector is considered here.

Definition 3.1 (Calibrated asymptotically Minkowskian MS end). *A reconstructed MS configuration belongs to the calibrated asymptotically Minkowskian phase space $\Gamma_\infty^{\text{MS}}$ if, relative to the calibrated reference response $\bar{\Theta}$, its canonical data satisfy*

$$h_{ij} = \delta_{ij} + \frac{a_{ij}(\hat{x})}{r} + O_2(r^{-2}), \quad (79)$$

$$\pi^{ij} = \frac{b^{ij}(\hat{x})}{r^2} + O_1(r^{-3}), \quad (80)$$

with Regge–Teitelboim parity

$$a_{ij}(-\hat{x}) = a_{ij}(\hat{x}), \quad (81)$$

$$b^{ij}(-\hat{x}) = -b^{ij}(\hat{x}). \quad (82)$$

Admissible tangent variations preserve these falloff and parity conditions, as well as the response-calibration and constraint-integrability conditions specified below.

The parity conditions in Eqs. (81)–(82) serve their standard canonical purpose. The potentially logarithmic part of the symplectic potential behaves asymptotically as

$$\int^\infty \frac{dr}{r} \int_{S^2} b^{ij}(\hat{x}) \delta a_{ij}(\hat{x}) d\Omega. \quad (83)$$

Since b^{ij} is odd and δa_{ij} is even, the angular integral vanishes. The canonical symplectic form

$$\Omega_\Sigma = \int_\Sigma d^3x \delta\pi^{ij} \wedge \delta h_{ij} \quad (84)$$

is therefore finite on the adopted phase space. This is the usual Regge–Teitelboim mechanism [7, 2], not an additional MS-specific result.

Using Eq. (77), the momentum falloff is equivalently expressed as

$$K_{ij} = \frac{k_{ij}(\hat{x})}{r^2} + O_1(r^{-3}), \quad k_{ij}(-\hat{x}) = -k_{ij}(\hat{x}). \quad (85)$$

Thus the leading configuration deviation appears at order r^{-1} , whereas the leading hypersurface-normal boost data appear at order r^{-2} . The distinction will later become the response-side origin of the different asymptotic orders supporting energy and spatial momentum.

3.2 Response calibration and the asymptotic translation sector

The local construction of Section 2 separates causal structure from absolute scale. A positive response rescaling

$$\Theta \mapsto \Theta' = \lambda \Theta, \quad \lambda > 0, \quad (86)$$

induces

$$g \mapsto g' = \lambda^2 g. \quad (87)$$

Such a transformation preserves the causal cone but changes the calibrated metric and therefore cannot in general be treated as gauge in a dimensionful asymptotic charge problem.

The relevant asymptotic orders can already be identified before any boundary charge is calculated. Suppose

$$\lambda = 1 + \frac{\sigma(\hat{x})}{r} + o(r^{-1}). \quad (88)$$

Then, to leading order,

$$h'_{ij} = h_{ij} + \frac{2\sigma(\hat{x})}{r} \delta_{ij} + o(r^{-1}), \quad (89)$$

and hence

$$a'_{ij} = a_{ij} + 2\sigma(\hat{x})\delta_{ij}. \quad (90)$$

If the transformed data are to remain in the same Regge–Teitelboim parity class, then

$$\sigma(-\hat{x}) = \sigma(\hat{x}). \quad (91)$$

Parity preservation, however, does not make this rescaling gauge: it changes the leading asymptotic metric data.

The normal first derivative of the response scale acts independently on the boost sector. Under $g' = \lambda^2 g$, Eq. (76) gives

$$K'_{ij} = \lambda \left[K_{ij} + h_{ij} n(\ln \lambda) \right], \quad (92)$$

where n is the future unit normal with respect to g . If

$$n(\ln \lambda) = \frac{\rho(\hat{x})}{r^2} + o(r^{-2}), \quad (93)$$

then the leading r^{-2} boost data are changed. Preservation of Eq. (85) requires

$$\rho(-\hat{x}) = -\rho(\hat{x}). \quad (94)$$

Equations (88) and (93) show that the response normalization has independent configuration and normal-jet data precisely at the orders where the asymptotic translation charges can be supported. We therefore impose the following sufficient calibration condition:

$$\lambda = 1 + o(r^{-1}), \quad \bar{D}_i \lambda = o(r^{-2}), \quad n(\ln \lambda) = o(r^{-2}). \quad (95)$$

We refer to Eq. (95) as *first-jet asymptotic response calibration*. It removes residual response-scale freedom at the orders relevant to the translation problem. It is a convenient sufficient condition, not a claim of mathematical minimality. After the translation charge has been constructed, we shall return to weaker charge-preserving conditions and identify which angular modes of σ and ρ are actually visible to the four translation charges.

It is important to keep three transformations conceptually distinct:

$$\begin{aligned} \text{internal Lorentz gauge : } & e \mapsto \Lambda e, \quad g \mapsto g, \\ \text{response recalibration : } & \Theta \mapsto \lambda \Theta, \quad g \mapsto \lambda^2 g, \\ \text{asymptotic inertial change : } & a \mapsto \Lambda_\infty a. \end{aligned} \tag{96}$$

The first is an internal redundancy of the soldered description. The second changes the physical metric scale unless it is below the calibrated orders. The third, introduced more fully after the total charge has been constructed, changes the inertial representation of the same asymptotic translation structure. Conflating these operations would obscure the distinction between gauge, calibration, and observer covariance.

The Regge–Teitelboim falloff controls the asymptotic symplectic structure but does not by itself guarantee absolute integrability of a bulk constraint generator smeared by a nonzero constant lapse or shift. For the off-shell boundary analysis we therefore restrict to the sufficient class

$$\mathcal{H}_\perp = O(r^{-3-\epsilon}), \quad \mathcal{H}_i = O(r^{-3-\epsilon}), \quad \epsilon > 0, \tag{97}$$

with admissible tangent variations satisfying

$$\delta \mathcal{H}_\perp = O(r^{-3-\epsilon}), \quad \delta \mathcal{H}_i = O(r^{-3-\epsilon}). \tag{98}$$

These are technical sufficient assumptions defining the off-shell domain used in the present paper; no claim is made that they are the weakest possible conditions. On the physical constraint surface,

$$\mathcal{H}_\perp \approx 0, \quad \mathcal{H}_i \approx 0. \tag{99}$$

The distinction matters because functional differentiability is a property of a neighborhood in phase space, not merely of the on-shell values of the constraints.

We further restrict to asymptotic deformation generators whose Hamiltonian vector fields are tangent to $\Gamma_\infty^{\text{MS}}$, including the falloff, parity, calibration, and constraint-integrability classes above. This condition will be assumed whenever an asymptotic Hamiltonian flow is used below.

An admissible asymptotic translation representative has lapse and shift

$$N = a^\perp + \nu, \quad \nu = O_1(r^{-1}), \tag{100}$$

$$\beta^i = a^i + \nu^i, \quad \nu^i = O_1(r^{-1}), \tag{101}$$

where

$$a^\perp \in \mathbb{R}, \quad a^i \in \mathbb{R}^3 \tag{102}$$

are constant relative to the calibrated reference end. Two representatives define the same asymptotic translation if they differ by a deformation whose lapse and shift vanish at infinity. Quotienting by this proper sector gives

$$\mathfrak{t}_\infty \simeq \mathbb{R} \oplus \mathbb{R}^3, \tag{103}$$

and we denote its elements by

$$a = (a^\perp, a^i) \in \mathfrak{t}_\infty. \tag{104}$$

This is the asymptotic counterpart of the translation space whose dual carries the relativistic generator in Ref. [3]. At this stage no total translation covector has yet been defined; only the space on which such a covector would act has been identified.

The physical criterion selecting a genuine asymptotic translation can now be stated.

Principle 3.2 (Asymptotic observer–translation principle). *On a calibrated asymptotically Minkowskian MS end, a deformation whose lapse and shift approach $a \in \mathfrak{t}_\infty$ represents a physical asymptotic observer translation only if its canonical generator is finite, integrable in field space, and functionally differentiable on $\Gamma_\infty^{\text{MS}}$.*

This principle does not reconstruct the bulk constraints a second time. Those constraints belong to the restricted classical completion of Ref. [5]. The principle instead asks whether their compact-support Hamiltonian realization survives when the deformation is allowed to approach a nonzero physical translation at infinity.

The answer will be negative for the bare bulk generator. Its variation contains a nonvanishing surface obstruction. The next section determines the unique boundary completion, up to the usual phase-space constant, that removes this obstruction. That completion will define the total translation covector

$$\mathbf{P}_\infty^{\text{MS}} \in \mathfrak{t}_\infty^*. \quad (105)$$

The logical output of the present section may therefore be summarized as

$$\overline{\Theta} \longrightarrow \text{calibrated asymptotic geometry} \longrightarrow \Gamma_\infty^{\text{MS}} \longrightarrow \mathfrak{t}_\infty. \quad (106)$$

The first arrow fixes the asymptotic physical scale, the second defines the canonical isolated-system phase space, and the third separates proper asymptotic gauge from the four-dimensional translation sector. The passage

$$\mathfrak{t}_\infty \longrightarrow \mathfrak{t}_\infty^*$$

is not made until functional differentiability supplies the required boundary Hamiltonian.

4 Boundary Completion and the Total Translation Covector

Section 3 identified the four-dimensional asymptotic translation space

$$\mathfrak{t}_\infty \simeq \mathbb{R} \oplus \mathbb{R}^3,$$

but did not assign a physical charge to its elements. The missing step is Hamiltonian differentiability.

The bulk deformation generators entering the present analysis are not introduced anew. They belong to the restricted classical gravitational completion of Ref. [5], in which the locally reconstructed soldering geometry yields the canonical variables (h_{ij}, π^{ij}) and the standard hypersurface-deformation constraint densities. In the compact-support sector used there, surface terms could be discarded. They cannot be discarded once the lapse and shift approach nonzero asymptotic translations.

The boundary mechanism used below is the standard Hamiltonian mechanism of asymptotically flat canonical gravity [7, 2]. No new ADM surface functional is claimed. The manuscript-specific statement is instead logical: the generator that requires completion has already arisen from the MS state–response–soldering construction and its restricted classical closure.

We continue to use $c = 1$. The bulk constraint densities are

$$\mathcal{H}_\perp = \frac{2\kappa}{\sqrt{h}} \left(\pi^{ij} \pi_{ij} - \frac{1}{2} \pi^2 \right) - \frac{\sqrt{h}}{2\kappa} {}^{(3)}R, \quad (107)$$

$$\mathcal{H}_i = -2D_j \pi^j_i, \quad (108)$$

where

$$\pi := h_{ij}\pi^{ij}. \quad (109)$$

For a lapse–shift pair $\chi = (N, \beta^i)$, the bare bulk generator is

$$G_0[\chi] := \int_{\Sigma} d^3x \left(N\mathcal{H}_{\perp} + \beta^i \mathcal{H}_i \right). \quad (110)$$

For compactly supported χ , this is the generator already used in the preceding bulk construction. We now allow

$$N = a^{\perp} + O_1(r^{-1}), \quad (111)$$

$$\beta^i = a^i + O_1(r^{-1}), \quad (112)$$

with

$$a = (a^{\perp}, a^i) \in \mathfrak{t}_{\infty}. \quad (113)$$

The constraint-integrability conditions of Section 3 make the bulk integral finite. Finiteness, however, is weaker than functional differentiability.

4.1 Differentiability obstruction and its boundary primitive

Let

$$\Sigma_R := \Sigma \cap \{r \leq R\}, \quad S_R := \partial\Sigma_R. \quad (114)$$

The limit $R \rightarrow \infty$ is taken only after the variation has been evaluated at finite radius. We denote by $\bar{s}^i$ and $dS_{\bar{h}}$ the outward unit normal and area element of S_R with respect to the calibrated Euclidean reference metric.

The kinetic part of $\mathcal{H}_{\perp}$ is ultralocal in the canonical variables and produces no derivative surface term. The normal-sector obstruction therefore comes entirely from

$$V_R[N] := -\frac{1}{2\kappa} \int_{\Sigma_R} d^3x N \sqrt{h} {}^{(3)}R. \quad (115)$$

Its finite-radius variation has the form

$$\delta V_R[N] = (\delta V_R[N])_{\text{bulk}} + \mathcal{S}_R^{\perp}[N; \delta h], \quad (116)$$

where

$$\mathcal{S}_R^{\perp}[N; \delta h] = -\frac{1}{2\kappa} \oint_{S_R} dS_{\bar{h}} \bar{s}^i \left[N \left(D^j \delta h_{ij} - D_i \delta h \right) - (D^j N) \delta h_{ij} + (D_i N) \delta h \right], \quad (117)$$

with

$$\delta h := h^{ij} \delta h_{ij}. \quad (118)$$

A finite-radius derivation of Eq. (117) may be placed in Appendix A.

For an asymptotic translation,

$$D_i N = O(r^{-2}),$$

while admissible variations satisfy

$$\delta h_{ij} = O(r^{-1}), \quad \bar{D}_k \delta h_{ij} = O(r^{-2}).$$

Using

$$D - \bar{D} = O(r^{-2}), \quad s^i - \bar{s}^i = O(r^{-1}),$$

one finds that all contributions involving $D_i N$, the subleading part of N , or the difference between physical and reference surface geometry vanish in the limit. Since

$$q_{ij} = h_{ij} - \bar{h}_{ij}$$

and the calibrated reference is fixed in phase space, the surviving normal obstruction is

$$\mathcal{S}_R^\perp[N; \delta h] = -a^\perp \delta E_R^{\text{MS}} + o(1), \quad (119)$$

where

$$E_R^{\text{MS}} := \frac{1}{2\kappa} \oint_{S_R} (\bar{D}^j q_{ij} - \bar{D}_i q^j{}_j) \bar{s}^i dS_{\bar{h}}. \quad (120)$$

The limit

$$E_\infty^{\text{MS}} := \frac{1}{2\kappa} \lim_{R \rightarrow \infty} \oint_{S_R} (\bar{D}^j q_{ij} - \bar{D}_i q^j{}_j) \bar{s}^i dS_{\bar{h}} \quad (121)$$

is finite under the falloff of Section 3. Thus the normal boundary obstruction is integrable in field space, and its primitive is

$$B_\infty^\perp[a] = a^\perp E_\infty^{\text{MS}}. \quad (122)$$

The tangential sector is similarly controlled by the momentum constraint. At finite radius,

$$D_R[\beta] := -2 \int_{\Sigma_R} d^3x \beta^i D_j \pi^j{}_i. \quad (123)$$

Because π^{ij} is a symmetric contravariant tensor density of weight +1, integration by parts gives

$$D_R[\beta] = \int_{\Sigma_R} d^3x \pi^{ij} \mathcal{L}_\beta h_{ij} - 2 \oint_{S_R} \beta^i \pi_i{}^j dS_j, \quad (124)$$

where

$$dS_j := \frac{1}{2} \epsilon_{jkl} dx^k \wedge dx^l \quad (125)$$

is the oriented coordinate surface element appropriate to a weight-one vector density. It is related to the geometric surface element by

$$\sqrt{h} dS_j = s_j dS_h. \quad (126)$$

The exact variation of Eq. (124) contains the variation of the density flux together with the finite-radius metric remainder

$$\mathcal{R}_R^\parallel[\beta; \delta h] = \oint_{S_R} \beta^k \pi^{ij} \delta h_{ij} dS_k. \quad (127)$$

The adopted falloff gives

$$\mathcal{R}_R^\parallel = O(R^{-1}). \quad (128)$$

Writing

$$\beta^i = a^i + v^i, \quad v^i = O(r^{-1}),$$

the subleading term also contributes only $O(R^{-1})$. Consequently, the only surviving tangential boundary variation is cancelled by the variation of

$$a^i J_{i,R}^{\text{MS}}, \quad J_{i,R}^{\text{MS}} := 2 \oint_{S_R} \pi_i^j dS_j. \quad (129)$$

The asymptotic spatial translation coefficient is therefore

$$J_i^{\text{MS}} := 2 \lim_{R \rightarrow \infty} \oint_{S_R} \pi_i^j dS_j. \quad (130)$$

Using Eq. (77) and Eq. (126), this may equivalently be written as

$$J_i^{\text{MS}} = \frac{1}{K} \lim_{R \rightarrow \infty} \oint_{S_R} (K^j_i - K \delta^j_i) s_j dS_h. \quad (131)$$

No additional factor of $\sqrt{h}$ is present in Eq. (131). The quantity J_i^{MS} is, at this stage, the Hamiltonian coefficient dual to a spatial asymptotic translation. Its relation to the conventional physical three-momentum will be fixed only after units and the sign convention of Eq. (76) are restored.

The full boundary primitive is therefore

$$B_\infty[a] := a^\perp E_\infty^{\text{MS}} + a^i J_i^{\text{MS}}. \quad (132)$$

For any admissible representative $\chi_a = (N, \beta^i)$ of $a \in \mathfrak{t}_\infty$, define

$$G[a] := G_0[\chi_a] + B_\infty[a]. \quad (133)$$

Theorem 4.1 (Boundary completion of the reconstructed translation generator). *Let $\Gamma_\infty^{\text{MS}}$ be the calibrated asymptotically Minkowskian phase space of Section 3, including its falloff, parity, first-jet response calibration, constraint-integrability, and linearized constraint-integrability conditions. Let $a \in \mathfrak{t}_\infty$.*

Then the bare reconstructed bulk generator $G_0[\chi_a]$ is finite but, for a nonzero asymptotic translation, is not in general functionally differentiable. Its surface obstruction is the exact field-space variation

$$-\delta B_\infty[a],$$

where $B_\infty[a]$ is given by Eq. (132). Consequently,

$$G[a] = G_0[\chi_a] + B_\infty[a]$$

is finite and functionally differentiable on $\Gamma_\infty^{\text{MS}}$.

Proof. Finiteness of the bulk integral follows from Eq. (97) and the bounded leading lapse and shift.

In the normal sector, the kinetic part of $\mathcal{H}_\perp$ is ultralocal and produces no derivative boundary term. The curvature term gives Eq. (117); after the asymptotic expansion, its only finite contribution is $-a^\perp \delta E_\infty^{\text{MS}}$.

In the tangential sector, Eq. (124) shows that the only finite translation contribution is the variation of the leading momentum-density flux. The metric remainder Eq. (127) and all terms involving the subleading shift decay as $O(R^{-1})$. The surviving term is therefore $-a^i \delta J_i^{\text{MS}}$.

Combining the two sectors gives

$$\delta G_0[\chi_a] = (\delta G_0[\chi_a])_{\text{bulk}} - \delta B_\infty[a]. \quad (134)$$

Hence $\delta G[a]$ contains no residual surface variation, and the corresponding Hamiltonian vector field is well defined on the stated asymptotic phase space. $\square$

4.2 The translation-dual charge

Functional differentiability determines the boundary completion up to a field-independent constant. Indeed, suppose $\widetilde{B}_\infty[a]$ is another boundary functional for the same asymptotic translation class such that

$$G_0[\chi_a] + \widetilde{B}_\infty[a]$$

is differentiable on the same phase space. Then

$$\delta(\widetilde{B}_\infty[a] - B_\infty[a]) = 0. \quad (135)$$

On each connected component,

$$\widetilde{B}_\infty[a] - B_\infty[a] = C[a], \quad (136)$$

where $C[a]$ is independent of the dynamical fields.

The calibrated Minkowski response $\overline{\Theta}$ selects a distinguished vacuum configuration

$$h_{ij} = \delta_{ij}, \quad \pi^{ij} = 0, \quad (137)$$

and we fix the origin of the charge space by requiring

$$B_\infty[a]|_{\overline{\Theta}} = 0 \quad \text{for all } a \in \mathfrak{t}_\infty. \quad (138)$$

This removes the residual constant.

Corollary 4.2 (Uniqueness after vacuum calibration). *For fixed $\Gamma_\infty^{\text{MS}}$, fixed calibrated reference response $\overline{\Theta}$, and fixed $a \in \mathfrak{t}_\infty$, the translation-sector boundary completion satisfying Eq. (138) is unique.*

Because Eq. (132) is linear in the asymptotic translation parameter, it defines a canonical dual element.

Definition 4.3 (Total asymptotic MS translation covector). *The total asymptotic MS translation covector is*

$$\mathbf{P}_\infty^{\text{MS}} \in \mathfrak{t}_\infty^* \quad (139)$$

defined by

$$\mathbf{P}_\infty^{\text{MS}}(a) := B_\infty[a] = a^\perp E_\infty^{\text{MS}} + a^i J_i^{\text{MS}}. \quad (140)$$

If $\{e_0, e_i\}$ is the translation basis determined by the calibrated asymptotic inertial frame and $\{e^0, e^i\}$ its dual basis, then in the present $c = 1$ convention

$$\mathbf{P}_\infty^{\text{MS}} = E_\infty^{\text{MS}} e^0 + J_i^{\text{MS}} e^i. \quad (141)$$

This equation is a Hamiltonian duality statement. It should not yet be read as a componentwise identification with the conventional positive-energy four-momentum vector. The conversion from the energy-valued translation covector to the physical momentum vector, including the factors of c and the spatial sign determined by our extrinsic-curvature convention, will be carried out only after the observer-space geometry of the charge has been established.

On the physical constraint surface,

$$\mathcal{H}_\perp \approx 0, \quad \mathcal{H}_i \approx 0, \quad (142)$$

and hence

$$G[a] \approx B_\infty[a] = P_\infty^{\text{MS}}(a). \quad (143)$$

The vanishing of the bulk constraint generator therefore does not imply the vanishing of the physical Hamiltonian associated with an improper asymptotic translation. Proper deformations have zero asymptotic translation class and carry no translation charge; nontrivial $a \in \mathfrak{t}_\infty$ can retain a finite boundary Hamiltonian.

This is the precise point at which the construction passes from an asymptotic symmetry parameter to a physical global generator:

$$\mathfrak{t}_\infty \longrightarrow P_\infty^{\text{MS}} \in \mathfrak{t}_\infty^*. \quad (144)$$

The structure parallels, without identifying, the translation duality of the one-particle theory in Ref. [3]. There the relativistic particle generator is a covector dual to flat translations; here the total generator of an isolated gravitating system becomes a covector dual to asymptotic translations only after the bulk geometry has been reconstructed, an asymptotic calibration has been chosen, and the Hamiltonian has been completed at infinity.

The two covectors must not be conflated. In particular, no sum of local one-particle massive scales is used in Eq. (140). The total gravitational charge is a functional of the complete asymptotic canonical geometry. Generic curved spacetimes provide no canonical path-independent identification of local cotangent spaces, and gravitational binding prevents the invariant mass of an isolated system from being represented by a naive sum of local positive massive generators.

The dependence on the preceding operational construction is therefore structural rather than local-density based:

$$\begin{aligned} \text{massive operational response} &\longrightarrow g[\Theta] \\ &\longrightarrow (h_{ij}, \pi^{ij}) \\ &\longrightarrow G_0[a] \\ &\longrightarrow B_\infty[a] \\ &\longrightarrow P_\infty^{\text{MS}} \in \mathfrak{t}_\infty^*. \end{aligned} \quad (145)$$

There is no local gravitational energy density implied by this chain. The local response reconstructs the geometry; the total translation generator appears only after that geometry is placed in the calibrated asymptotically flat Hamiltonian sector.

The existence of P_∞^{MS} does not yet establish all of its physical properties. In particular, functional differentiability alone does not determine whether the reduced translation generators commute, how the covector transforms under a change of asymptotic inertial representation, or under which admissible evolutions it is conserved. These questions will be addressed in the next section before the Lorentzian observer-space geometry and invariant mass of the total charge are analyzed.

5 Translation Symmetry, Covariance, and Conservation

Section 4 associates with every $a \in \mathfrak{t}_\infty$ a differentiable Hamiltonian

$$G[a] = G_0[\chi_a] + B_\infty[a]$$

and hence a total translation covector

$$P_\infty^{\text{MS}} \in \mathfrak{t}_\infty^*, \quad P_\infty^{\text{MS}}(a) = B_\infty[a].$$

Three further properties are required before this object can be interpreted as the relativistic total momentum of the isolated system. First, the translation charges must realize the Abelian asymptotic translation algebra after proper deformations are removed. Second, the four components must assemble into a single Lorentz-covariant object under changes of asymptotic inertial representation. Third, the resulting charge must be conserved under admissible physical evolution.

These issues are distinct from the boundary-completion problem itself. In particular, we shall not require a global off-shell central-extension analysis of the full hypersurface-deformation algebra. For the translation sector it is sufficient, and mathematically cleaner, to work directly with the asymptotic action on the boundary data and then pass to the reduced phase space.

5.1 Reduced asymptotic translation algebra

For every

$$a = (a^\perp, a^i) \in \mathfrak{t}_\infty,$$

choose an admissible representative

$$\chi_a = (N_a, \beta_a^i) \quad (146)$$

that agrees outside a compact set with the constant asymptotic deformation

$$N_a = a^\perp, \quad \beta_a^i = a^i. \quad (147)$$

Different interior extensions represent the same asymptotic translation whenever their difference has vanishing leading lapse and shift.

The surface-deformation bracket of two such representatives has no constant translation component. Indeed, in the asymptotic region

$$D_i N_a = D_i N_b = 0, \quad [\beta_a, \beta_b]^i = 0. \quad (148)$$

Consequently, the normal and tangential components of their surface-deformation bracket vanish asymptotically,

$$[\chi_a, \chi_b]_{\text{SD}} = \chi_{ab}^{\text{proper}}, \quad (149)$$

where $\chi_{ab}^{\text{proper}}$ has zero asymptotic translation class. Thus the quotient algebra is

$$[a, b]_\infty = 0, \quad \mathfrak{t}_\infty \simeq \mathbb{R}^4 \quad (150)$$

as an Abelian Lie algebra.

What remains is to verify that the boundary charges themselves carry no residual translation-sector bracket. This can be established directly from their asymptotic variations.

Let $b \in \mathfrak{t}_\infty$ generate an admissible canonical translation. Hamilton's equation for the spatial metric is

$$\delta_b h_{ij} = \frac{4\kappa N_b}{\sqrt{h}} \left(\pi_{ij} - \frac{1}{2} \pi h_{ij} \right) + \mathcal{L}_{\beta_b} h_{ij}. \quad (151)$$

Using

$$h_{ij} = \delta_{ij} + O_2(r^{-1}), \quad \pi^{ij} = O_1(r^{-2}),$$

and the fact that N_b and the leading coordinate components of β_b^i are constant near infinity, one obtains

$$\delta_b h_{ij} = O(r^{-2}). \quad (152)$$

The momentum equation gives correspondingly

$$\delta_b \pi^{ij} = O(r^{-3}). \quad (153)$$

The leading curvature contribution has this order because ${}^{(3)}G^{ij} = O(r^{-3})$; the terms quadratic in π^{ij} decay faster, and the Lie derivative along the constant asymptotic shift differentiates the leading r^{-2} momentum data.

These estimates immediately determine the variation of the boundary coefficients. From Eq. (121),

$$\begin{aligned} \delta_b E_\infty^{\text{MS}} &= \frac{1}{2\kappa} \lim_{R \rightarrow \infty} \oint_{S_R} (\bar{D}^j \delta_b h_{ij} - \bar{D}_i \delta_b h^j_j) \bar{s}^i dS_{\bar{h}} \\ &= 0, \end{aligned} \quad (154)$$

because the differentiated variation is $O(r^{-3})$ while the sphere area grows as $O(r^2)$.

Likewise, Eq. (130) gives

$$\delta_b J_i^{\text{MS}} = 2 \lim_{R \rightarrow \infty} \oint_{S_R} \delta_b \pi_i^j dS_j = 0. \quad (155)$$

The additional contribution produced when the metric used to lower the index of π^{ij} is varied is $O(r^{-4})$ and does not affect the limit. Hence

$$\delta_b B_\infty[a] = 0 \quad \text{for all } a, b \in \mathfrak{t}_\infty. \quad (156)$$

On the constraint surface,

$$G[a] \approx B_\infty[a] = \mathbf{P}_\infty^{\text{MS}}(a),$$

while a proper deformation represents the zero element after quotienting by proper asymptotic gauge. Equation (156) therefore gives the required reduced algebra directly.

Theorem 5.1 (Reduced asymptotic translation algebra). *Let $\Gamma_\infty^{\text{MS}}$ be the calibrated asymptotic phase space of Section 3, and assume that the admissible translation flows preserve this phase space.*

For any $a, b \in \mathfrak{t}_\infty$, their surface-deformation bracket has vanishing asymptotic translation class, and the associated total boundary charges satisfy

$$\{\mathbf{P}_\infty^{\text{MS}}(a), \mathbf{P}_\infty^{\text{MS}}(b)\}_{\text{red}} = 0. \quad (157)$$

Thus the translation charges furnish an ordinary Abelian Hamiltonian representation of $\mathfrak{t}_\infty \simeq \mathbb{R}^4$ on the reduced phase space.

Proof. Equation (149) shows that the bracket of two translation representatives differs from zero only by a proper asymptotic deformation. Such a deformation represents zero on the reduced phase space.

For the boundary part, Eq. (156) shows directly that the Hamiltonian flow generated by any asymptotic translation leaves every translation boundary functional invariant. Therefore

$$\delta_b \mathbf{P}_\infty^{\text{MS}}(a) = 0.$$

With the reduced Hamiltonian action this is precisely Eq. (157). $\square$

The theorem is intentionally a reduced statement. It is sufficient for the physical translation sector considered here and avoids attributing an ordinary Lie-algebra structure to the full field-dependent hypersurface-deformation bracket. No claim is made about possible central extensions of enlarged asymptotic symmetry algebras containing boosts, rotations, or angle-dependent transformations.

5.2 Lorentz covariance and conservation

The calibrated flat reference response equips the asymptotic translation space with a Lorentzian bilinear form

$$\eta_\infty : \mathfrak{t}_\infty \times \mathfrak{t}_\infty \longrightarrow \mathbb{R}, \quad \text{sign}(\eta_\infty) = (-, +, +, +). \quad (158)$$

Choose a future-oriented orthonormal translation basis

$$\{e_0, e_i\}, \quad \eta_\infty(e_0, e_0) = -1, \quad \eta_\infty(e_i, e_j) = \delta_{ij}, \quad (159)$$

with dual basis $\{e^0, e^i\}$.

The boundary functional constructed in Section 4 has components

$$\mathbf{P}_\infty^{\text{MS}} = E_\infty^{\text{MS}} e^0 + J_i^{\text{MS}} e^i. \quad (160)$$

Its transformation law follows directly from the invariance of the scalar Hamiltonian pairing.

Let

$$\Lambda \in SO^+(1, 3)$$

relate two asymptotic inertial representations, with translation labels related by

$$a' = \Lambda a. \quad (161)$$

The geometric Hamiltonian associated with a translation does not depend on the basis used to represent that translation, so

$$\mathbf{P}_\infty^{\text{MS}}(a) = \mathbf{P}_\infty^{\text{MS}}(a'). \quad (162)$$

Consequently,

$$\mathbf{P}_\infty^{\prime \text{MS}} = (\Lambda^{-1})^* \mathbf{P}_\infty^{\text{MS}}, \quad (163)$$

or, in matrix notation,

$$\mathbf{P}_\infty^{\prime \text{MS}} = \Lambda^{-T} \mathbf{P}_\infty^{\text{MS}}.$$

At this point it is useful to connect the boundary notation with the vector–covector convention used in Refs. [5, 4]. The covector $\mathbf{P}_\infty^{\text{MS}}$ constructed by the boundary Hamiltonian is naturally positive on a future time translation. It therefore plays the role of a *positive-response covector*. Define

$$\widehat{\Pi}_\infty^{\text{MS}} := \mathbf{P}_\infty^{\text{MS}}, \quad (164)$$

and introduce the canonical action covector

$$\Pi_\infty^{\text{MS}} := -\widehat{\Pi}_\infty^{\text{MS}}. \quad (165)$$

The associated positive-energy vector is

$$P_\infty^{\text{MS}} := -\eta_\infty^\#(\widehat{\Pi}_\infty^{\text{MS}}) = \eta_\infty^\#(\Pi_\infty^{\text{MS}}). \quad (166)$$

This is precisely the vector–covector polarity used in the local soldering theory.

In the basis Eq. (159), Eq. (160) therefore gives

$$P_\infty^{\text{MS}} = E_\infty^{\text{MS}} e_0 - J_i^{\text{MS}} e_i. \quad (167)$$

Define

$$p_{\infty i} := -J_i^{\text{MS}}. \quad (168)$$

Then, in the present $c = 1$ convention,

$$P_{\infty}^{\text{MS}} = (E_{\infty}^{\text{MS}}, \mathbf{p}_{\infty}), \quad (169)$$

whereas

$$\Pi_{\infty}^{\text{MS}} = (-E_{\infty}^{\text{MS}}, \mathbf{p}_{\infty}). \quad (170)$$

Thus the global convention now agrees exactly with the local convention

$$P^a = (E, \mathbf{p}), \quad \Pi_a = (-E, \mathbf{p})$$

used in the preceding soldering and quantum papers.

Because Λ preserves η_{∞} , Eq. (163) implies

$$P_{\infty}^{\prime \text{MS}} = \Lambda P_{\infty}^{\text{MS}}. \quad (171)$$

Proposition 5.2 (Asymptotic Lorentz covariance). *The total boundary charge is a covector on $\mathfrak{t}_{\infty}$, while its Lorentz-polar positive-energy representative P_{∞}^{MS} is a vector. Under a change of calibrated asymptotic inertial representation,*

$$\widehat{\Pi}_{\infty}^{\text{MS}} \mapsto (\Lambda^{-1})^* \widehat{\Pi}_{\infty}^{\text{MS}}, \quad (172)$$

$$P_{\infty}^{\text{MS}} \mapsto \Lambda P_{\infty}^{\text{MS}}. \quad (173)$$

This covariance statement does not construct the asymptotic Lorentz generators themselves. Rotations and boosts require asymptotic lapse or shift fields with linear growth and additional boundary analysis. The present result states only that the already constructed four-dimensional translation space carries the Lorentz representation inherited from the calibrated Minkowski reference.

The distinction among the three transformations introduced in Section 3 is now particularly sharp:

$$\begin{aligned} \text{internal Lorentz gauge : } & (g, P_{\infty}^{\text{MS}}) \text{ unchanged,} \\ \text{asymptotic inertial change : } & \widehat{\Pi}_{\infty}^{\text{MS}} \mapsto (\Lambda^{-1})^* \widehat{\Pi}_{\infty}^{\text{MS}}, \\ & P_{\infty}^{\text{MS}} \mapsto \Lambda P_{\infty}^{\text{MS}}, \\ \text{response recalibration : } & g \mapsto \lambda^2 g, \end{aligned} \quad (174)$$

with the last transformation generically changing the boundary charge unless it lies below the charge-supporting asymptotic orders.

We finally establish conservation. Let the physical evolution be generated by an admissible Hamiltonian whose asymptotic translation class is a fixed element

$$t \in \mathfrak{t}_{\infty}, \quad (175)$$

and assume, as in Section 3, that the corresponding Hamiltonian flow preserves $\Gamma_{\infty}^{\text{MS}}$. For every time-independent $a \in \mathfrak{t}_{\infty}$, the reduced evolution of the charge is

$$\frac{d}{dt} P_{\infty}^{\text{MS}}(a) = \{P_{\infty}^{\text{MS}}(a), P_{\infty}^{\text{MS}}(t)\}_{\text{red}}. \quad (176)$$

Theorem 5.1 therefore gives

$$\frac{d}{dt} \mathbf{P}_\infty^{\text{MS}}(a) = 0 \quad \text{for every fixed } a \in \mathfrak{t}_\infty. \quad (177)$$

Hence

$$\frac{d}{dt} \mathbf{P}_\infty^{\text{MS}} = 0 \quad (178)$$

on the reduced physical phase space. Since the asymptotic Lorentzian form η_∞ is fixed by the calibrated reference, Eq. (166) also implies

$$\frac{d}{dt} P_\infty^{\text{MS}} = 0. \quad (179)$$

Corollary 5.3 (Conservation of the total translation charge). *For admissible physical evolution preserving the calibrated asymptotic phase space and approaching a fixed asymptotic translation, both the total translation covector $\mathbf{P}_\infty^{\text{MS}}$ and its Lorentz-polar positive-energy vector P_∞^{MS} are conserved on the reduced phase space.*

The construction has therefore reached a complete classical translation sector:

$$\mathbf{P}_\infty^{\text{MS}} \in \mathfrak{t}_\infty^*, \quad \left\{ \mathbf{P}_\infty^{\text{MS}}(a), \mathbf{P}_\infty^{\text{MS}}(b) \right\}_{\text{red}} = 0, \quad \dot{\mathbf{P}}_\infty^{\text{MS}} = 0. \quad (180)$$

No invariant mass has been assumed in obtaining these statements. The next section asks what Lorentz-invariant scalar and what distinguished observer are selected once the conserved total translation vector P_∞^{MS} is probed over the future unit observer hyperboloid. This will provide the global counterpart of the local response polarity developed in Section 2.

6 Observer-Space Geometry of Total Momentum

The preceding sections constructed the total translation charge before introducing any notion of invariant mass. We now have

$$\widehat{\Pi}_\infty^{\text{MS}} = \mathbf{P}_\infty^{\text{MS}} \in \mathfrak{t}_\infty^* \quad (181)$$

and the associated positive-energy vector

$$P_\infty^{\text{MS}} = -\eta_\infty^\sharp(\widehat{\Pi}_\infty^{\text{MS}}) \in \mathfrak{t}_\infty. \quad (182)$$

The present section asks how much of this global generator can be recovered from the scalar energy assigned to it by asymptotic inertial observers.

The Lorentzian and hyperbolic identities used below are standard. Their role here is not to provide a new definition of relativistic mass, but to give an observer-space readout of a charge that has already been obtained independently from the MS boundary construction. The manuscript-specific structure appears when this global observer problem is compared with the local massive-response polarity of Section 2.

We continue to use $c = 1$. Physical dimensions and the distinction between momentum-valued covectors, energy-valued boundary covectors, and physical invariant mass will be restored in Section 7.

6.1 Energy landscape and momentum reconstruction

Let

$$\mathcal{H}_\infty^+ := \{u \in \mathfrak{t}_\infty : \eta_\infty(u, u) = -1, \quad u \text{ future directed}\} \quad (183)$$

be the future unit observer hyperboloid, equipped with the induced Riemannian metric

$$h_H := \eta_\infty|_{T\mathcal{H}_\infty^+}. \quad (184)$$

For $u \in \mathcal{H}_\infty^+$, define the observer energy

$$\mathcal{E}_\infty(u) := \widehat{\Pi}_\infty^{\text{MS}}(u) = -\eta_\infty(P_\infty^{\text{MS}}, u). \quad (185)$$

Thus the completed boundary covector may equivalently be represented by the scalar function

$$\mathcal{E}_\infty : \mathcal{H}_\infty^+ \longrightarrow \mathbb{R}. \quad (186)$$

Choose a future-oriented orthonormal basis $\{e_0, e_i\}$ for $(\mathfrak{t}_\infty, \eta_\infty)$. As established in Section 5,

$$P_\infty^{\text{MS}} = E_\infty^{\text{MS}} e_0 + p_\infty^i e_i. \quad (187)$$

Every $u \in \mathcal{H}_\infty^+$ may be parameterized by the compact MS variables as

$$u(D, S) = \Phi(D, S) = \frac{1}{S} (e_0 + D^i e_i), \quad (188)$$

where

$$|D|^2 + S^2 = 1, \quad S > 0. \quad (189)$$

Consequently,

$$\mathcal{E}_\infty(D, S) = \frac{E_\infty^{\text{MS}} - \mathbf{p}_\infty \cdot \mathbf{D}}{S}. \quad (190)$$

Equivalently, with

$$D = \tanh \chi \, n, \quad S = \text{sech } \chi, \quad |n| = 1, \quad (191)$$

one has

$$u = \cosh \chi \, e_0 + \sinh \chi \, n^i e_i \quad (192)$$

and

$$\mathcal{E}_\infty(\chi, n) = E_\infty^{\text{MS}} \cosh \chi - (\mathbf{p}_\infty \cdot n) \sinh \chi. \quad (193)$$

The energy function contains more information than its minimum. Because it is the restriction to $\mathcal{H}_\infty^+$ of an ambient Lorentzian linear functional, its first derivative already recovers the full translation vector.

Proposition 6.1 (Observer-space reconstruction of total momentum). *Let $P_\infty^{\text{MS}} \in \mathfrak{t}_\infty$ be arbitrary and let $\mathcal{E}_\infty$ be defined by Eq. (185). Then, at every $u \in \mathcal{H}_\infty^+$,*

$$\nabla_H \mathcal{E}_\infty = -P_\infty^{\text{MS}} + \mathcal{E}_\infty u, \quad (194)$$

and therefore

$$P_\infty^{\text{MS}} = \mathcal{E}_\infty u - \nabla_H \mathcal{E}_\infty. \quad (195)$$

Moreover,

$$-\eta_\infty(P_\infty^{\text{MS}}, P_\infty^{\text{MS}}) = \mathcal{E}_\infty^2 - |\nabla_H \mathcal{E}_\infty|_{h_H}^2, \quad (196)$$

and

$$\text{Hess}_H \mathcal{E}_\infty = \mathcal{E}_\infty h_H. \quad (197)$$

In 3 + 1 spacetime dimensions,

$$\Delta_H \mathcal{E}_\infty = 3\mathcal{E}_\infty. \quad (198)$$

Proof. For $v \in T_u \mathcal{H}_\infty^+$, one has $\eta_\infty(u, v) = 0$. The vector

$$-P_\infty^{\text{MS}} + \mathcal{E}_\infty u$$

is tangent because

$$\eta_\infty(-P_\infty^{\text{MS}} + \mathcal{E}_\infty u, u) = 0.$$

Furthermore,

$$h_H(-P_\infty^{\text{MS}} + \mathcal{E}_\infty u, v) = -\eta_\infty(P_\infty^{\text{MS}}, v) = d\mathcal{E}_\infty(v),$$

which proves Eq. (194). Equation (195) follows immediately.

Taking the Lorentzian norm of Eq. (195) and using

$$\eta_\infty(u, u) = -1, \quad \eta_\infty(u, \nabla_H \mathcal{E}_\infty) = 0$$

gives Eq. (196).

Finally, differentiating Eq. (194) in an observer-space tangent direction and projecting tangentially gives

$$\nabla_v^H \nabla_H \mathcal{E}_\infty = \mathcal{E}_\infty v.$$

Hence Eq. (197). Taking its trace over the three-dimensional observer hyperboloid yields Eq. (198). $\square$

Equation (195) may be read operationally as follows: at any asymptotic observer, the measured total energy together with its first variation under infinitesimal changes of observer determines the complete total translation vector. This interpretation is useful here, but the underlying hyperbolic identity is standard and is not claimed as a new reconstruction theorem.

The scalar on the left-hand side of Eq. (196) is independent of u . This constancy also follows intrinsically from Eq. (197), since

$$\begin{aligned} d(\mathcal{E}_\infty^2 - |\nabla_H \mathcal{E}_\infty|_{h_H}^2) &= 2\mathcal{E}_\infty d\mathcal{E}_\infty - 2 \text{Hess}_H \mathcal{E}_\infty (\nabla_H \mathcal{E}_\infty, \cdot) \\ &= 0. \end{aligned} \quad (199)$$

Thus the Lorentzian norm of the ambient translation vector is encoded entirely in the first-order geometry of the scalar energy landscape.

Suppose now that P_∞^{MS} is future timelike. Define

$$Q_\infty^{\text{MS}} := \sqrt{-\eta_\infty(P_\infty^{\text{MS}}, P_\infty^{\text{MS}})}. \quad (200)$$

The normalized vector

$$u_* := \frac{P_\infty^{\text{MS}}}{Q_\infty^{\text{MS}}} \in \mathcal{H}_\infty^+ \quad (201)$$

is the unique rest observer of the total charge.

If $\rho(u, u_*)$ denotes hyperbolic distance on $\mathcal{H}_\infty^+$, then

$$-\eta_\infty(u, u_*) = \cosh \rho(u, u_*), \quad (202)$$

so

$$\mathcal{E}_\infty(u) = Q_\infty^{\text{MS}} \cosh \rho(u, u_*). \quad (203)$$

Consequently,

$$Q_\infty^{\text{MS}} = \min_{u \in \mathcal{H}_\infty^+} \mathcal{E}_\infty(u), \quad (204)$$

and the minimum is attained uniquely at u_* .

At the rest observer,

$$\nabla_H \mathcal{E}_\infty|_{u_*} = 0 \quad (205)$$

and

$$\text{Hess}_H \mathcal{E}_\infty|_{u_*} = Q_\infty^{\text{MS}} h_H. \quad (206)$$

Thus the same radial invariant is simultaneously the minimum energy and the isotropic curvature scale of the observer-energy landscape at its minimum:

$$Q_\infty^{\text{MS}} = \min_{\mathcal{H}_\infty^+} \mathcal{E}_\infty = \frac{1}{3} \text{tr}_{h_H} [\text{Hess}_H \mathcal{E}_\infty]_{u_*}. \quad (207)$$

Since $\mathcal{E}_\infty \geq Q_\infty^{\text{MS}} > 0$, Eq. (197) also implies

$$\text{Hess}_H \mathcal{E}_\infty \geq Q_\infty^{\text{MS}} h_H. \quad (208)$$

The observer-energy landscape is therefore strictly geodesically convex in the future-timelike sector. Its unique minimum is the rest observer.

In compact MS coordinates, Eq. (201) becomes

$$D_\infty = \frac{\mathbf{p}_\infty}{E_\infty^{\text{MS}}}, \quad S_\infty = \frac{Q_\infty^{\text{MS}}}{E_\infty^{\text{MS}}}, \quad (209)$$

and hence

$$P_\infty^{\text{MS}} = Q_\infty^{\text{MS}} \Phi(D_\infty, S_\infty). \quad (210)$$

This is formally parallel to the one-particle projective factorization of Ref. [3], but the radial quantities have different physical meanings. The local shell scale Q belongs to an individual massive generator, whereas Q_∞^{MS} is the Lorentzian radial invariant of the boundary-completed total translation charge of the entire isolated configuration. No identification between the two is implied.

6.2 Causal spectrum and local–global response reciprocity

The boundary construction itself does not force P_∞^{MS} to be future timelike. Its complete Lorentzian causal classification can be read directly from the observer-energy function.

Proposition 6.2 (Observer-energy classification). *Let*

$$\mathcal{E}_\infty(u) = -\eta_\infty(P_\infty^{\text{MS}}, u), \quad u \in \mathcal{H}_\infty^+.$$

Then

$$\inf_{u \in \mathcal{H}_\infty^+} \mathcal{E}_\infty(u) = \begin{cases} Q_\infty^{\text{MS}} > 0, & P_\infty^{\text{MS}} \text{ future timelike,} \\ 0, & P_\infty^{\text{MS}} \text{ nonzero future null,} \\ 0, & P_\infty^{\text{MS}} = 0, \\ -\infty, & P_\infty^{\text{MS}} \text{ spacelike or nonzero past directed.} \end{cases} \quad (211)$$

The infimum is attained uniquely in the future-timelike case, is not attained for a nonzero future-null vector, is attained by every observer for the zero vector, and is unbounded below in the remaining sectors.

Proof. The future-timelike case follows from Eq. (203).

For a nonzero future-null vector, choose the spatial direction n parallel to $\mathbf{p}_\infty$. Then

$$E_\infty^{\text{MS}} = |\mathbf{p}_\infty|$$

and Eq. (193) gives

$$\mathcal{E}_\infty = |\mathbf{p}_\infty| e^{-\chi} \longrightarrow 0 \quad (\chi \rightarrow \infty). \quad (212)$$

No finite observer attains the value zero.

If $P_\infty^{\text{MS}} = 0$, then $\mathcal{E}_\infty \equiv 0$.

If P_∞^{MS} is spacelike, choose n parallel to $\mathbf{p}_\infty$. Since $|\mathbf{p}_\infty| > |E_\infty^{\text{MS}}|$, the leading large- χ coefficient in Eq. (193) is negative and

$$\mathcal{E}_\infty \longrightarrow -\infty.$$

The same conclusion holds for a nonzero past-directed timelike or null vector. $\square$

Thus the future causal cone of the total translation vector is characterized operationally by

$$P_\infty^{\text{MS}} \text{ future causal} \iff \mathcal{E}_\infty(u) \geq 0 \quad \text{for every } u \in \mathcal{H}_\infty^+. \quad (213)$$

Equivalently,

$$P_\infty^{\text{MS}} \text{ future causal} \iff \inf_{\mathcal{H}_\infty^+} \mathcal{E}_\infty \geq 0. \quad (214)$$

Strict positivity of the infimum is equivalent to future timelikeness. These statements are consequences of Lorentz-cone polarity; they are not a positive-mass theorem. Whether the physical gravitational data actually place P_∞^{MS} in the future causal cone requires additional dynamical hypotheses and will be discussed only after the ADM identification.

The nonzero future-null sector provides a direct global counterpart of the projective boundary encountered in the local massive theory. From Eq. (212), the minimizing sequence satisfies

$$|D| \longrightarrow 1, \quad S \longrightarrow 0^+. \quad (215)$$

Thus the compact MS boundary again represents an ideal infinite-rapidity limit. A nonzero null total momentum has no finite rest observer, just as a null direction is not itself a finite point of the massive observer hyperboloid.

We can now state the local–global structural relation that motivates the observer-space analysis.

Locally, Section 2 holds a spacetime displacement ξ fixed and varies the positive massive-generator covector:

$$\mu_x(\xi) = \frac{1}{Q} \inf_{\widehat{\Pi} \in \widehat{\Sigma}_{Q,x}^+} \Theta_x^+(\widehat{\Pi}, \xi) = \sqrt{-g_x(\xi, \xi)}. \quad (216)$$

Globally, once boundary completion has supplied a fixed total generator, one instead varies the future unit observer:

$$Q_\infty^{\text{MS}} = \inf_{u \in \mathcal{H}_\infty^+} \widehat{\Pi}_\infty^{\text{MS}}(u) = \sqrt{-\eta_\infty(P_\infty^{\text{MS}}, P_\infty^{\text{MS}})} \quad (217)$$

in the future-timelike sector.

Proposition 6.3 (Local–global response reciprocity). *Within the MS construction, the local and global variational problems are two realizations of the same Lorentzian polarity with the roles of probe and generator exchanged:*

$$\begin{aligned} \text{local: } & \text{fixed displacement} + \text{variable massive generator} \\ & \longrightarrow \text{local proper-time norm,} \\ \text{global: } & \text{fixed total generator} + \text{variable asymptotic observer} \\ & \longrightarrow \text{global radial invariant.} \end{aligned} \quad (218)$$

The proposition does not assert a new Lorentzian inequality. Its content is architectural: the two polar problems occur at opposite ends of the same operational construction. The local generator shell enters before the spacetime metric has been reconstructed, whereas the global observer hyperboloid enters only after the reconstructed geometry has acquired a boundary-completed total translation generator.

The two radial quantities must also remain distinct:

$$Q \text{ labels a local one-particle massive shell,} \quad Q_\infty^{\text{MS}} \text{ measures the total boundary generator.} \quad (219)$$

In particular, no relation of the form

$$Q_\infty^{\text{MS}} = \sum_A Q_A$$

is implied. The global generator is a boundary functional of the complete asymptotic gravitational geometry, not a sum of local massive generators.

There is nevertheless a common projective structure. For a future-timelike total charge,

$$P_\infty^{\text{MS}} = Q_\infty^{\text{MS}} \Phi(D_\infty, S_\infty) \quad (220)$$

separates the projective rest-observer direction from the radial magnitude. The same distinction already appears in the one-particle theory: the compact state determines a hyperbolic direction, whereas the shell parameter supplies its radial scale.

This separation will become physically important in the next section. At fixed asymptotic geometry, a common change in the gravitational normalization rescales the radial boundary charge but does not alter its projective rest-observer direction. By contrast, an asymptotic recalibration of the operational response changes the geometric data themselves and can therefore change the charge. The next section distinguishes these two effects, identifies the charge-supporting response modes, restores physical units, and compares the resulting translation charge with the standard ADM energy–momentum.

7 Response Calibration and ADM Identification

The total translation charge has so far been constructed without using the ADM four-momentum as a definition. The preceding sections required two different kinds of normalization that must now be separated carefully.

The first is the asymptotic calibration of the operational response. A transformation

$$\Theta \mapsto \lambda \Theta$$

changes the reconstructed geometry according to

$$g \mapsto \lambda^2 g$$

and can therefore modify the boundary data themselves. The second is the common gravitational coefficient κ appearing in the canonical generator. Once the asymptotic geometry is held fixed, κ controls the overall magnitude assigned to the gravitational translation charge.

These two operations have different logical status:

$$\begin{aligned} \text{response calibration} &\longrightarrow \text{absolute asymptotic geometry,} \\ \text{gravitational normalization} &\longrightarrow \text{radial magnitude of its translation charge.} \end{aligned} \quad (221)$$

The distinction is central to the projective–radial separation already encountered in Section 6.

7.1 Charge-supporting response modes

We first return to the response rescaling considered in Section 3,

$$\Theta \mapsto \Theta' = \lambda \Theta, \quad \lambda > 0, \quad (222)$$

with

$$g' = \lambda^2 g. \quad (223)$$

The strong first-jet calibration imposed in the definition of $\Gamma_\infty^{\text{MS}}$,

$$\lambda = 1 + o(r^{-1}), \quad \bar{D}_i \lambda = o(r^{-2}), \quad n(\ln \lambda) = o(r^{-2}), \quad (224)$$

removes all response-scale ambiguity at the asymptotic orders capable of supporting the four translation charges. It is useful, however, to relax this condition temporarily and determine which modes are actually seen by those charges.

Suppose first that

$$\lambda = 1 + \frac{\sigma(\hat{x})}{r} + o(r^{-1}). \quad (225)$$

To leading order,

$$\Delta_\lambda h_{ij} = \frac{2\sigma(\hat{x})}{r} \delta_{ij} + o(r^{-1}). \quad (226)$$

Thus the leading Regge–Teitelboim coefficient transforms as

$$a_{ij} \mapsto a'_{ij} = a_{ij} + 2\sigma(\hat{x})\delta_{ij}. \quad (227)$$

Preservation of the adopted parity class requires

$$\sigma(-\hat{x}) = \sigma(\hat{x}). \quad (228)$$

Let

$$f(x) := \frac{\sigma(\hat{x})}{r}. \quad (229)$$

Using Eq. (121), the induced change in the temporal translation coefficient is

$$\begin{aligned} \Delta_\lambda E_\infty^{\text{MS}} &= \frac{1}{2\kappa} \lim_{R \rightarrow \infty} \oint_{S_R} [\partial^j (2f \delta_{ij}) - \partial_i (6f)] \hat{x}^i dS \\ &= -\frac{2}{\kappa} \lim_{R \rightarrow \infty} \oint_{S_R} \hat{x}^i \partial_i f dS. \end{aligned} \quad (230)$$

Since

$$\hat{x}^i \partial_i \left(\frac{\sigma(\hat{x})}{r} \right) = -\frac{\sigma(\hat{x})}{r^2}, \quad (231)$$

one obtains

$$\Delta_\lambda E_\infty^{\text{MS}} = \frac{2}{\kappa} \int_{S^2} \sigma(\hat{x}) d\Omega. \quad (232)$$

Therefore only the spherical monopole of the leading r^{-1} response-scale mode contributes to the total energy.

The normal first jet of the response scale acts independently on the momentum-supporting sector. Under $g' = \lambda^2 g$, the extrinsic curvature transforms, with the convention Eq. (76), as

$$K'_{ij} = \lambda [K_{ij} + h_{ij} n(\ln \lambda)]. \quad (233)$$

Consequently,

$$K'^j_i - K' \delta^j_i = \lambda^{-1} [K^j_i - K \delta^j_i - 2n(\ln \lambda) \delta^j_i]. \quad (234)$$

Suppose

$$n(\ln \lambda) = \frac{\rho(\hat{x})}{r^2} + o(r^{-2}). \quad (235)$$

Preservation of the Regge–Teitelboim parity class requires

$$\rho(-\hat{x}) = -\rho(\hat{x}). \quad (236)$$

The conformal factors multiplying the pre-existing $K_{ij} = O(r^{-2})$ terms differ from unity only at sub-leading order and do not contribute a finite shift. Equation (131) therefore gives

$$\Delta_\lambda J_i^{\text{MS}} = -\frac{2}{\kappa} \int_{S^2} \rho(\hat{x}) \hat{x}_i d\Omega. \quad (237)$$

Since $\hat{x}_i$ carries the $l = 1$ spherical representation, only the $l = 1$ projection of the allowed odd function ρ contributes to the three spatial translation coefficients.

We may summarize the translation-sensitive response data as

$$\begin{aligned} \lambda - 1 &= \frac{\sigma(\hat{x})}{r} + \dots, \quad \sigma_{l=0} \longrightarrow E_\infty^{\text{MS}}, \\ n(\ln \lambda) &= \frac{\rho(\hat{x})}{r^2} + \dots, \quad \rho_{l=1} \longrightarrow J_i^{\text{MS}}. \end{aligned} \quad (238)$$

Proposition 7.1 (Translation-sensitive response calibration). *Within the Regge–Teitelboim translation sector, a response recalibration preserving the asymptotic parity class changes the four translation coefficients only through*

$$\Delta_\lambda E_\infty^{\text{MS}} = \frac{2}{\kappa} \int_{S^2} \sigma d\Omega, \quad \Delta_\lambda J_i^{\text{MS}} = -\frac{2}{\kappa} \int_{S^2} \rho \hat{x}_i d\Omega. \quad (239)$$

Thus the temporal charge reads only the monopole of the leading spatial response recalibration, while the spatial translation charges read only the dipole of its normal first jet.

The spherical-harmonic statement itself is standard: relative to a chosen asymptotic observer, the four-dimensional translation representation decomposes into an $SO(3)$ scalar and vector sector. The MS-specific content of Proposition 7.1 is the identification of those sectors with two distinct pieces of the operational response calibration: the leading spatial scale and its normal first jet.

The strong condition Eq. (224) is sufficient to remove both effects pointwise. If one asks only for the four integrated translation charges to remain unchanged, the weaker leading-order conditions

$$\int_{S^2} \sigma(\hat{x}) d\Omega = 0, \quad \int_{S^2} \rho(\hat{x}) \hat{x}_i d\Omega = 0 \quad (240)$$

are sufficient at the orders displayed above. Higher even modes of σ and higher odd modes of ρ are invisible to the four translation charges, although they need not be irrelevant to higher asymptotic moments or to enlarged asymptotic symmetry sectors.

At the translation level, the charge map therefore acts as a finite symmetry-selected projection of the asymptotic response jet:

$$(\sigma, \rho) \mapsto \left(\int_{S^2} \sigma d\Omega, \int_{S^2} \rho \hat{x}_i d\Omega \right) \in \mathbb{R} \oplus \mathbb{R}^3. \quad (241)$$

This projection should not be interpreted as a complete multipole theory. It identifies only the response data visible to the asymptotic translation sector treated in the present paper.

There is a second scale dependence that must be kept separate. Hold the asymptotic geometric data (h_{ij}, K_{ij}) fixed and compare two positive choices of the common canonical coefficient κ . Equations (121) and (131) give

$$\mathbf{P}_\infty^{\text{MS}} \propto \kappa^{-1}. \quad (242)$$

Hence

$$\kappa \mapsto \alpha \kappa, \quad \alpha > 0, \quad (243)$$

implies, at fixed geometry,

$$E_\infty^{\text{MS}} \mapsto \alpha^{-1} E_\infty^{\text{MS}}, \quad \mathbf{p}_\infty \mapsto \alpha^{-1} \mathbf{p}_\infty, \quad Q_\infty^{\text{MS}} \mapsto \alpha^{-1} Q_\infty^{\text{MS}}. \quad (244)$$

The compact rest-observer coordinates, however,

$$D_\infty = \frac{\mathbf{p}_\infty}{E_\infty^{\text{MS}}}, \quad S_\infty = \frac{Q_\infty^{\text{MS}}}{E_\infty^{\text{MS}}}, \quad (245)$$

are unchanged.

Thus a common gravitational normalization changes the radial magnitude of the total translation charge but not its projective direction at fixed geometry. This comparison is not a gauge transformation within one fixed phase space. It compares different common normalizations of the same

geometric canonical data. By contrast, the response recalibrations in Eqs. (225) and (235) change the asymptotic geometry itself.

The hierarchy is therefore

$$\begin{aligned}
&\text{internal Lorentz frame} \longrightarrow \text{gauge representative,} \\
&\text{asymptotic inertial frame} \longrightarrow \text{components of the same charge,} \\
&\text{response calibration} \longrightarrow \text{absolute asymptotic geometry,} \\
&\kappa \longrightarrow \text{radial gravitational charge normalization.}
\end{aligned} \tag{246}$$

7.2 Physical units and Einstein-normalized ADM realization

We now restore physical units. This step requires care because several objects that coincide numerically when $c = 1$ carry different dimensions.

The boundary functional

$$\mathbf{P}_\infty^{\text{MS}} \in \mathfrak{t}_\infty^*$$

is energy valued:

$$\mathbf{P}_\infty^{\text{MS}} = E_\infty^{\text{MS}} e^0 - c P_{\infty i}^{\text{MS}} e^i. \tag{247}$$

Here $\mathbf{P}_\infty^{\text{MS}}$ denotes the physical three-momentum, defined by

$$J_i^{\text{MS}} = -c P_{\infty i}^{\text{MS}}. \tag{248}$$

The momentum-valued positive-response covector is therefore

$$\widehat{\Pi}_\infty^{\text{MS}} := \frac{1}{c} \mathbf{P}_\infty^{\text{MS}} = \frac{E_\infty^{\text{MS}}}{c} e^0 - P_{\infty i}^{\text{MS}} e^i, \tag{249}$$

while the canonical action covector is

$$\Pi_\infty^{\text{MS}} := -\widehat{\Pi}_\infty^{\text{MS}} = -\frac{E_\infty^{\text{MS}}}{c} e^0 + P_{\infty i}^{\text{MS}} e^i. \tag{250}$$

The positive-energy four-momentum vector is

$$P_\infty^{\text{MS}} := -\eta_\infty^\#(\widehat{\Pi}_\infty^{\text{MS}}) = \frac{E_\infty^{\text{MS}}}{c} e_0 + P_{\infty i}^{\text{MS}} e_i. \tag{251}$$

Thus

$$P_\infty^{\text{MS}} = \left(\frac{E_\infty^{\text{MS}}}{c}, \mathbf{P}_\infty^{\text{MS}} \right), \quad \Pi_\infty^{\text{MS}} = \left(-\frac{E_\infty^{\text{MS}}}{c}, \mathbf{P}_\infty^{\text{MS}} \right), \tag{252}$$

in exact agreement with the convention used in Refs. [5, 4].

The equality $\widehat{\Pi}_\infty^{\text{MS}} = \mathbf{P}_\infty^{\text{MS}}$ used in the preceding $c = 1$ sections was therefore numerical only. Dimensionally,

$$\mathbf{P}_\infty^{\text{MS}} = c \widehat{\Pi}_\infty^{\text{MS}}. \tag{253}$$

The restricted classical closure fixes the relative normalization of the kinetic and curvature terms but does not determine the numerical gravitational coupling. To compare with physical general relativity we now impose the Einstein normalization

$$\kappa = \frac{8\pi G}{c^4}. \tag{254}$$

This is an empirical normalization of the gravitational response; it is not derived from the compact-state relation, the operational soldering theorem, or the restricted canonical closure.

Equation (121) then becomes

$$E_{\infty}^{\text{MS}} = \frac{c^4}{16\pi G} \lim_{R \rightarrow \infty} \oint_{S_R} (\bar{D}^j q_{ij} - \bar{D}_i q^j{}_j) \bar{s}^i dS_{\bar{h}}. \quad (255)$$

This is the standard ADM energy surface functional.

For the spatial component, Eqs. (131) and (248) give

$$P_{\infty i}^{\text{MS}} = -\frac{1}{cK} \lim_{R \rightarrow \infty} \oint_{S_R} (K^j{}_i - K \delta^j{}_i) s_j dS_h, \quad (256)$$

or

$$P_{\infty i}^{\text{MS}} = -\frac{c^3}{8\pi G} \lim_{R \rightarrow \infty} \oint_{S_R} (K^j{}_i - K \delta^j{}_i) s_j dS_h. \quad (257)$$

Our extrinsic-curvature convention is

$$K_{ij} = \frac{1}{2N} (\dot{h}_{ij} - \mathcal{L}_{\beta} h_{ij}).$$

If the conventional ADM sign

$$K_{ij}^{\text{ADM}} = -K_{ij} \quad (258)$$

is used instead, then Eq. (257) becomes

$$P_{\infty i}^{\text{MS}} = \frac{c^3}{8\pi G} \lim_{R \rightarrow \infty} \oint_{S_R} (K^{\text{ADM} j}{}_i - K^{\text{ADM}} \delta^j{}_i) s_j dS_h, \quad (259)$$

which is the standard ADM momentum expression.

Theorem 7.2 (Einstein-normalized ADM realization). *Consider the calibrated asymptotically Minkowskian phase space of Section 3 in the minimal torsion-free metric branch of the preceding MS reconstruction. After the empirical normalization*

$$\kappa = \frac{8\pi G}{c^4}$$

is imposed, the energy and physical spatial momentum obtained from the MS boundary completion satisfy

$$E_{\infty}^{\text{MS}} = E_{\text{ADM}}, \quad \mathbf{P}_{\infty}^{\text{MS}} = \mathbf{P}_{\text{ADM}}. \quad (260)$$

Consequently,

$$P_{\infty}^{\text{MS}} = P_{\text{ADM}} = \left(\frac{E_{\text{ADM}}}{c}, \mathbf{P}_{\text{ADM}} \right). \quad (261)$$

Proof. The temporal equality follows from

$$\frac{1}{2\kappa} = \frac{c^4}{16\pi G},$$

which turns the differentiability boundary term of the reconstructed scalar generator into Eq. (255).

In the spatial sector,

$$\frac{1}{cK} = \frac{c^3}{8\pi G}.$$

Together with $J_i^{\text{MS}} = -cP_{\infty i}^{\text{MS}}$ and the sign conversion $K_{ij}^{\text{ADM}} = -K_{ij}$, this yields Eq. (259). The vector equality then follows from Eq. (251). $\square$

The theorem is an identification, not a new derivation of the ADM surface functional from canonical general relativity. The Regge–Teitelboim boundary mechanism and the resulting ADM charges are classical [7, 1, 2]. The MS-specific logical order is

$$\begin{aligned}
&\text{compact massive-state geometry} \longrightarrow \text{operational response} \\
&\quad \longrightarrow \text{soldered Lorentzian geometry} \\
&\quad \longrightarrow \text{restricted canonical bulk generator} \\
&\quad \longrightarrow \text{differentiable asymptotic boundary charge} \\
&\quad \longrightarrow P_{\infty}^{\text{MS}} \\
&\quad \xrightarrow{\kappa=8\pi G/c^4} P_{\text{ADM}}.
\end{aligned} \tag{262}$$

The standard ADM expression therefore appears as the Einstein-normalized realization of a translation charge whose operational provenance was fixed before the ADM comparison was made.

For a future-timelike total momentum, define

$$Q_{\infty}^{\text{MS}} := \sqrt{-\eta_{\infty}(P_{\infty}^{\text{MS}}, P_{\infty}^{\text{MS}})} = \sqrt{\left(\frac{E_{\infty}^{\text{MS}}}{c}\right)^2 - |\mathbf{P}_{\infty}^{\text{MS}}|^2}. \tag{263}$$

This quantity has momentum dimension. The corresponding physical invariant mass is

$$M_{\infty}^{\text{MS}} := \frac{Q_{\infty}^{\text{MS}}}{c} = \frac{1}{c^2} \sqrt{(E_{\infty}^{\text{MS}})^2 - c^2 |\mathbf{P}_{\infty}^{\text{MS}}|^2}. \tag{264}$$

Theorem 7.2 therefore gives

$$M_{\infty}^{\text{MS}} = M_{\text{ADM}}. \tag{265}$$

The observer-space result of Section 6 now acquires its physical form. For

$$u(D, S) = \frac{1}{S} (e_0 + D^i e_i), \quad |D|^2 + S^2 = 1, \quad S > 0, \tag{266}$$

the physical energy assigned to the total charge is

$$\begin{aligned}
E_{\text{obs}}(D, S) &:= P_{\infty}^{\text{MS}}(u(D, S)) \\
&= \frac{E_{\infty}^{\text{MS}} - c \mathbf{P}_{\infty}^{\text{MS}} \cdot \mathbf{D}}{S}.
\end{aligned} \tag{267}$$

Whenever the total four-momentum is future timelike,

$$M_{\infty}^{\text{MS}} c^2 = \min_{u \in \mathcal{H}_{\infty}^+} E_{\text{obs}}(u). \tag{268}$$

After the ADM identification,

$$M_{\text{ADM}} c^2 = \min_{|D|^2 + S^2 = 1, S > 0} \frac{E_{\text{ADM}} - c \mathbf{P}_{\text{ADM}} \cdot \mathbf{D}}{S}. \tag{269}$$

The minimizing observer is

$$D_{\infty} = \frac{c \mathbf{P}_{\text{ADM}}}{E_{\text{ADM}}}, \quad S_{\infty} = \frac{M_{\text{ADM}} c^2}{E_{\text{ADM}}}, \tag{270}$$

and the corresponding projective factorization is

$$P_{\text{ADM}} = M_{\text{ADM}} c \Phi(D_{\infty}, S_{\infty}). \quad (271)$$

Equation (271) is formally parallel to the one-particle relation

$$P = Q \Phi(D, S), \quad Q = mc, \quad (272)$$

but the radial quantities belong to different physical levels:

$$\frac{Q}{c} \quad \text{is a local one-particle rest mass,} \quad M_{\text{ADM}} \quad \text{is the invariant mass of the global gravitational charge.} \quad (273)$$

No identification or additive relation between them is implied.

Finally, the causal classification of Section 6 must be distinguished from a positive-mass theorem. The present construction allows one to classify an arbitrary completed translation charge as timelike, null, zero, spacelike, or past directed. It does not prove that physical asymptotically flat initial data necessarily produce a future-causal charge. After Theorem 7.2, the standard positive-mass theorem may be invoked under its own hypotheses [8, 9]. It then constrains the ADM vector to the future causal cone.

Under the corresponding rigidity hypotheses, the equality case is the flat one; in the usual asymptotically flat setting this excludes a nonzero null ADM four-momentum [6]. Thus the nonzero future-null line in Proposition 6.2 belongs to the complete Lorentzian classification of possible translation vectors, but need not represent an additional physical ADM sector once the full positive-mass rigidity assumptions are imposed.

The classical construction of the present paper has therefore reached its endpoint:

$$\begin{aligned} \Theta &\longrightarrow g[\Theta] \longrightarrow (h_{ij}, \pi^{ij}) \\ &\longrightarrow P_{\infty}^{\text{MS}} \in \mathfrak{t}_{\infty}^* \\ &\longrightarrow P_{\infty}^{\text{MS}} \\ &\longrightarrow (D_{\infty}, S_{\infty}, M_{\infty}^{\text{MS}}) \\ &\xrightarrow{\kappa=8\pi G/c^4} (D_{\infty}, S_{\infty}, M_{\text{ADM}}). \end{aligned} \quad (274)$$

The ADM identification fixes the standard gravitational realization of the radial charge. The projective observer data and the local–global response polarity were already determined before this final normalization.

8 Discussion and Outlook

The construction developed in this paper addresses a specific gap between the local and global levels of the MS programme. The compact-state analysis of Ref. [3] selects the standard relativistic massive-generator geometry and its translation-dual structure under the stated dynamical and observer-completeness assumptions. The operational soldering construction of Ref. [5] then converts the local scalar generator–displacement response into a nondegenerate solder map and hence into Lorentzian spacetime geometry. Within the restricted classical branch considered there, the same geometry admits a canonical hypersurface-deformation description.

None of these local statements, however, supplies a total energy or momentum for an isolated gravitating system. Generic curved spacetimes possess no global Abelian translation group, and local generator covectors at different events cannot be added without additional geometric identification. The present paper therefore addresses a genuinely new level of the construction: the remnant of translation symmetry that survives at a calibrated asymptotically Minkowskian boundary.

The logical chain established here is

$$\begin{aligned}
&\text{compact massive-state geometry} \longrightarrow \text{operational generator–displacement response} \\
&\longrightarrow \text{soldered Lorentzian geometry} \\
&\longrightarrow \text{restricted canonical bulk generator} \\
&\longrightarrow \text{calibrated asymptotic translation sector} \tag{275} \\
&\longrightarrow \mathbf{P}_\infty^{\text{MS}} \in \mathfrak{t}_\infty^* \\
&\longrightarrow P_\infty^{\text{MS}} \\
&\longrightarrow \text{observer-space invariant mass.}
\end{aligned}$$

After the empirical Einstein normalization

$$\kappa = \frac{8\pi G}{c^4}$$

is imposed, this total translation charge coincides with the standard ADM four-momentum in the asymptotically flat metric sector considered here.

The significance of this result should be separated carefully from the standard ingredients used in its derivation. The Regge–Teitelboim boundary-completion mechanism is classical, as are the ADM surface functionals, the Lorentzian minimum-energy characterization of rest mass, and the hyperbolic identities obeyed by ambient Minkowski linear functionals [7, 1, 2]. The present work does not claim new versions of these results. Its contribution lies instead in the provenance and organization of the structures being completed and compared.

The first distinctive feature is the continuation from local operational geometry to a global symmetry charge. The canonical generator entering the boundary calculation is not taken as the starting point of the programme. It occurs only after the massive response has reconstructed a soldered Lorentzian geometry and that geometry has entered the restricted classical closure. Functional differentiability then promotes a nontrivial asymptotic deformation from a bulk gauge generator with a surface obstruction to a genuine Hamiltonian symmetry carrying the translation-dual charge

$$\mathbf{P}_\infty^{\text{MS}} \in \mathfrak{t}_\infty^*.$$

In this sense the boundary charge is the global continuation of an operationally reconstructed bulk theory rather than an independently postulated observable.

A second structural feature is the recurrence of Lorentzian polarity at the local and global ends of the construction. Locally,

$$\frac{1}{Q} \inf_{\widehat{\Pi} \in \widehat{\Sigma}_{Q,x}^+} \widehat{\Pi}(e_x \xi) = \sqrt{-g_x(\xi, \xi)} \tag{276}$$

holds a displacement fixed while the positive massive-generator shell is varied. Globally,

$$\inf_{u \in \mathcal{H}_\infty^+} \widehat{\Pi}_\infty^{\text{MS}}(u) = \sqrt{-\eta_\infty(P_\infty^{\text{MS}}, P_\infty^{\text{MS}})} \tag{277}$$

holds the completed total generator fixed while the asymptotic observer is varied, whenever the total momentum is future timelike. The underlying Lorentzian duality is standard; what is specific to the present framework is that the two variational problems occur at opposite ends of the same state–response–geometry–boundary construction.

This role exchange also clarifies why the local shell parameter and the global invariant mass must not be identified. The parameter

$$Q = mc$$

sets the radial scale of an individual local massive shell, whereas

$$M_\infty^{\text{MS}} = \frac{1}{c} \sqrt{-\eta_\infty(P_\infty^{\text{MS}}, P_\infty^{\text{MS}})}$$

is the radial invariant of the complete boundary-generated momentum of an isolated gravitating configuration. In the Einstein-normalized sector, the latter equals M_{ADM} . No sum rule relating the two levels is implied.

The observer-space representation makes this distinction particularly transparent. The scalar function

$$E_{\text{obs}}(u) = -c \eta_\infty(P_\infty^{\text{MS}}, u) \quad (278)$$

contains the complete four-momentum information. In units $c = 1$,

$$P_\infty^{\text{MS}} = E_{\text{obs}} u - \nabla_H E_{\text{obs}}, \quad (279)$$

while

$$-\eta_\infty(P_\infty^{\text{MS}}, P_\infty^{\text{MS}}) = E_{\text{obs}}^2 - |\nabla_H E_{\text{obs}}|_{h_H}^2. \quad (280)$$

For a future-timelike charge,

$$\text{Hess}_H E_{\text{obs}} = E_{\text{obs}} h_H \quad (281)$$

and the unique rest observer is the minimum of this strictly convex energy landscape. These identities are classical hyperbolic geometry, but in the present construction they provide a compact scalar readout of a global charge obtained independently from Hamiltonian boundary completion.

A third distinctive feature concerns absolute response calibration. Positive local rescaling of the operational response preserves the causal cone but changes the reconstructed metric scale. At spatial infinity, the translation charges do not depend on all such asymptotic response data. At the leading orders considered here,

$$\lambda - 1 = \frac{\sigma(\hat{x})}{r} + \dots \quad (282)$$

affects the total energy only through the monopole of σ , whereas

$$n(\ln \lambda) = \frac{\rho(\hat{x})}{r^2} + \dots \quad (283)$$

affects the spatial translation charge only through the dipole of ρ . Thus the translation sector performs the finite-dimensional projection

$$(\sigma, \rho) \mapsto \left(\int_{S^2} \sigma d\Omega, \int_{S^2} \rho \hat{x}_i d\Omega \right) \in \mathbb{R} \oplus \mathbb{R}^3. \quad (284)$$

The familiar scalar-plus-vector decomposition of translations is standard. The point specific to the MS formulation is that these four modes are identified as distinct pieces of the asymptotic operational response jet.

This calibration analysis also separates two notions of scale that would otherwise be easy to conflate. A response recalibration

$$\Theta \mapsto \lambda \Theta$$

changes the reconstructed geometry itself. By contrast, at fixed geometric data a common change of the gravitational coefficient rescales the total charge as

$$P_{\infty}^{\text{MS}} \propto \kappa^{-1} \quad (285)$$

without changing its projective rest-observer direction. Hence

$$\text{projective observer state} \quad \text{and} \quad \text{radial gravitational magnitude} \quad (286)$$

are structurally distinct. This is the global analogue of the separation between the compact massive-state direction and the one-particle shell scale in Ref. [3].

The resulting picture is therefore not a derivation of general relativity from the compact MS relation alone. Several independent inputs remain essential. The asymptotically flat Regge–Teitelboim phase space is adopted rather than derived. The gravitational dynamics used here belongs to the restricted torsion-free metric branch of the classical completion. The value

$$\kappa = \frac{8\pi G}{c^4}$$

is empirically imposed and is not fixed by the operational state geometry. Likewise, the future-causal character of the physical ADM momentum follows only after the usual hypotheses of positive-energy theorems are supplied; it is not a consequence of the compact-state relation itself.

These limitations identify the natural extensions of the present work. The first is the completion of the full asymptotic Poincaré sector. Rotations and boosts require linearly growing asymptotic deformation fields and additional parity-sensitive boundary terms. Constructing those charges would test whether the translation-dual structure obtained here extends naturally to angular momentum and center-of-mass data.

A second direction is to relax the Regge–Teitelboim translation sector. The higher spherical harmonics of the response calibration that are invisible to Eq. (284) may become relevant once angle-dependent asymptotic transformations are allowed. An enlarged BMS-type analysis would determine whether those higher response modes acquire a direct charge interpretation or remain pure calibration data.

A third direction concerns stationary geometries. Once a conserved asymptotic translation vector has been established, additional Killing structure permits comparison with Komar-type charges and with local scalar response spectra on the stationary Killing plane. This would connect the boundary-Hamiltonian construction developed here with the independent closed-flux characterization available in stationary vacuum gravity.

Finally, the global boundary problem has so far been treated entirely classically. Ref. [4] follows a different branch of the same local response construction by promoting soldering configurations and their coherent deformations to a canonical quantum algebra. No quantum boundary charge has been constructed here. Whether the classical asymptotic translation functional admits a natural operator or algebraic realization, and how such a realization would interact with the projective comparison sector, remain separate questions.

The role of the present paper within the sequence can therefore be stated concisely:

$$\begin{aligned}
&\text{Paper I :} && \text{massive-state and flat translation rigidity,} \\
&\text{Paper II :} && \text{operational soldering and restricted local gravity,} \\
&\text{Paper III :} && \text{quantum configuration and projective comparison,} \\
&\text{Paper IV :} && \text{classical global boundary completion and total momentum.}
\end{aligned} \tag{287}$$

The present result closes the classical local-to-global step: a scalar operational response first reconstructs local Lorentzian geometry; the resulting bulk gravitational generator then acquires a translation-dual boundary charge; and that global charge is again completely readable through a scalar response over asymptotic observer space.

This closure, rather than the ADM surface formula by itself, is the main structural content of the paper.

A Finite-Radius Boundary Variation

This appendix supplies the finite-radius variational calculation used in Section 4. Its purpose is limited: to verify that the surface obstruction of the reconstructed bulk generator is precisely the negative field-space variation of the translation boundary functional. The underlying Hamiltonian boundary mechanism is standard [7, 2].

Throughout this appendix $c = 1$. Let

$$\Sigma_R := \Sigma \cap \{r \leq R\}, \quad S_R := \partial \Sigma_R, \tag{288}$$

with R sufficiently large that S_R lies in the asymptotically Cartesian end. The physical outward unit normal and area element are denoted by

$$s^i, \quad dS_h,$$

while reference quantities associated with

$$\bar{h}_{ij} = \delta_{ij}$$

carry an overbar:

$$\bar{s}^i = \hat{x}^i, \quad dS_{\bar{h}} = R^2 d\Omega. \tag{289}$$

The asymptotic canonical data satisfy

$$h_{ij} = \delta_{ij} + O_2(r^{-1}), \quad \pi^{ij} = O_1(r^{-2}), \tag{290}$$

and admissible tangent variations obey

$$\delta h_{ij} = O_2(r^{-1}), \quad \delta \pi^{ij} = O_1(r^{-2}). \tag{291}$$

The canonical momentum π^{ij} is a symmetric contravariant tensor density of weight +1.

A.1 Normal and tangential surface terms

Consider first the curvature contribution to the scalar generator,

$$V_R[N] := -\frac{1}{2\kappa} \int_{\Sigma_R} d^3x N \sqrt{h} {}^{(3)}R. \quad (292)$$

For variations of the covariant metric,

$$\delta \sqrt{h} = \frac{1}{2} \sqrt{h} h^{ij} \delta h_{ij}, \quad (293)$$

and

$$\delta {}^{(3)}R = -{}^{(3)}R^{ij} \delta h_{ij} + D^i D^j \delta h_{ij} - D^2 \delta h, \quad \delta h := h^{ij} \delta h_{ij}. \quad (294)$$

Hence

$$\delta \left(\sqrt{h} {}^{(3)}R \right) = \sqrt{h} \left[-{}^{(3)}G^{ij} \delta h_{ij} + D^i D^j \delta h_{ij} - D^2 \delta h \right]. \quad (295)$$

After integrating the derivative terms by parts at finite radius,

$$\delta V_R[N] = (\delta V_R[N])_{\text{bulk}} + \mathcal{S}_R^\perp[N; \delta h], \quad (296)$$

where

$$\begin{aligned} \mathcal{S}_R^\perp[N; \delta h] = & -\frac{1}{2\kappa} \oint_{S_R} dS_h s^i \left[N \left(D^j \delta h_{ij} - D_i \delta h \right) \right. \\ & \left. - (D^j N) \delta h_{ij} + (D_i N) \delta h \right]. \end{aligned} \quad (297)$$

Write

$$q_{ij} := h_{ij} - \bar{h}_{ij}. \quad (298)$$

The physical and reference connections differ by

$$C^k_{ij} := \Gamma^k_{ij}[h] - \bar{\Gamma}^k_{ij} = O(r^{-2}), \quad (299)$$

so that

$$D_k \delta h_{ij} - \bar{D}_k \delta h_{ij} = O(r^{-3}). \quad (300)$$

Similarly,

$$s^i = \bar{s}^i + O(r^{-1}), \quad dS_h = \left(1 + O(r^{-1}) \right) dS_{\bar{h}}. \quad (301)$$

For an asymptotic time translation,

$$N = a^\perp + \nu, \quad \nu = O_1(r^{-1}), \quad (302)$$

and therefore

$$D_i N = O(r^{-2}). \quad (303)$$

The terms containing $D_i N$, the subleading lapse ν , and the difference between physical and reference surface geometry all decay as $O(R^{-1})$ after integration over S_R . Thus

$$\begin{aligned} \mathcal{S}_R^\perp[N; \delta h] = & -\frac{a^\perp}{2\kappa} \oint_{S_R} \left(\bar{D}^j \delta h_{ij} - \bar{D}_i \delta h^j_j \right) \bar{s}^i dS_{\bar{h}} \\ & + O(R^{-1}). \end{aligned} \quad (304)$$

Define the finite-radius energy functional

$$E_R^{\text{MS}} := \frac{1}{2\kappa} \oint_{S_R} (\bar{D}^j q_{ij} - \bar{D}_i q^j{}_j) \bar{s}^i dS_{\bar{h}}. \quad (305)$$

Since the reference metric is fixed,

$$\delta q_{ij} = \delta h_{ij},$$

and therefore

$$\mathcal{S}_R^\perp[N; \delta h] + \delta(a^\perp E_R^{\text{MS}}) = O(R^{-1}). \quad (306)$$

The kinetic part of the scalar constraint,

$$T_R[N] := 2\kappa \int_{\Sigma_R} d^3x \frac{N}{\sqrt{h}} \left(\pi^{ij} \pi_{ij} - \frac{1}{2} \pi^2 \right), \quad (307)$$

contains no derivatives of the canonical variations and hence produces no surface variation. Its asymptotic bulk density is $O(r^{-4})$, so the corresponding radial integral converges.

We next consider the tangential generator,

$$D_R[\beta] := -2 \int_{\Sigma_R} d^3x \beta^i D_j \pi^j{}_i. \quad (308)$$

Because π^{ij} is a symmetric contravariant density of weight +1,

$$D_R[\beta] = \int_{\Sigma_R} d^3x \pi^{ij} \mathcal{L}_\beta h_{ij} - 2 \oint_{S_R} \beta^i \pi_i{}^j dS_j, \quad (309)$$

where

$$dS_j := \frac{1}{2} \epsilon_{jkl} dx^k \wedge dx^l \quad (310)$$

is the oriented coordinate surface element.

The variation of the bulk Lie-derivative term is

$$\begin{aligned} \delta \int_{\Sigma_R} d^3x \pi^{ij} \mathcal{L}_\beta h_{ij} &= \int_{\Sigma_R} d^3x \left[(\mathcal{L}_\beta h_{ij}) \delta \pi^{ij} - (\mathcal{L}_\beta \pi^{ij}) \delta h_{ij} \right] \\ &\quad + \oint_{S_R} \beta^k \pi^{ij} \delta h_{ij} dS_k. \end{aligned} \quad (311)$$

Hence

$$\begin{aligned} \delta D_R[\beta] &= \int_{\Sigma_R} d^3x \left[(\mathcal{L}_\beta h_{ij}) \delta \pi^{ij} - (\mathcal{L}_\beta \pi^{ij}) \delta h_{ij} \right] \\ &\quad - 2\delta \oint_{S_R} \beta^i \pi_i{}^j dS_j + \mathcal{R}_R^\parallel, \end{aligned} \quad (312)$$

where

$$\mathcal{R}_R^\parallel := \oint_{S_R} \beta^k \pi^{ij} \delta h_{ij} dS_k. \quad (313)$$

For an asymptotic spatial translation,

$$\beta^i = a^i + v^i, \quad v^i = O_1(r^{-1}), \quad (314)$$

the remainder satisfies

$$\mathcal{R}_R^\parallel = O(R^{-1}). \quad (315)$$

The subleading shift likewise gives

$$\oint_{S_R} \nu^i \pi_i^j dS_j = O(R^{-1}). \quad (316)$$

Define

$$J_{i,R}^{\text{MS}} := 2 \oint_{S_R} \pi_i^j dS_j. \quad (317)$$

Then

$$-2\delta \oint_{S_R} \beta^i \pi_i^j dS_j + \delta(a^i J_{i,R}^{\text{MS}}) = O(R^{-1}), \quad (318)$$

and therefore

$$\begin{aligned} \delta D_R[\beta] + \delta(a^i J_{i,R}^{\text{MS}}) &= \int_{\Sigma_R} d^3x \left[(\mathcal{L}_\beta h_{ij}) \delta \pi^{ij} \right. \\ &\quad \left. - (\mathcal{L}_\beta \pi^{ij}) \delta h_{ij} \right] + O(R^{-1}). \end{aligned} \quad (319)$$

The two finite-radius boundary coefficients have finite limits:

$$E_\infty^{\text{MS}} = \lim_{R \rightarrow \infty} E_R^{\text{MS}}, \quad J_i^{\text{MS}} = \lim_{R \rightarrow \infty} J_{i,R}^{\text{MS}}. \quad (320)$$

Their elementary power-counting finiteness follows from

$$\bar{D}q = O(r^{-2}), \quad \pi = O(r^{-2}).$$

For later comparison with the geometric form of the spatial charge, the weight-one density satisfies

$$\pi^{ij} = \frac{\sqrt{h}}{2K} (K^{ij} - K h^{ij}). \quad (321)$$

The coordinate and geometric surface elements obey

$$dS_j = \frac{s_j}{\sqrt{h}} dS_h, \quad (322)$$

and hence

$$J_i^{\text{MS}} = \frac{1}{K} \lim_{R \rightarrow \infty} \oint_{S_R} (K^j_i - K \delta^j_i) s_j dS_h. \quad (323)$$

No additional factor of $\sqrt{h}$ occurs.

A.2 Combined differentiability

Define the finite-radius improved generator

$$G_R[N, \beta] := \int_{\Sigma_R} d^3x \left(N \mathcal{H}_\perp + \beta^i \mathcal{H}_i \right) + a^\perp E_R^{\text{MS}} + a^i J_{i,R}^{\text{MS}}. \quad (324)$$

Combining Eqs. (306) and (319) gives

$$\delta G_R[N, \beta] = \int_{\Sigma_R} d^3x \left(\mathcal{A}^{ij}[N, \beta] \delta h_{ij} + \mathcal{B}_{ij}[N, \beta] \delta \pi^{ij} \right) + O(R^{-1}), \quad (325)$$

where $\mathcal{A}^{ij}$ and $\mathcal{B}_{ij}$ are the ordinary bulk Hamiltonian functional derivatives.

The off-shell asymptotic domain used in the main text additionally imposes

$$\mathcal{H}_\perp = O(r^{-3-\epsilon}), \quad \mathcal{H}_i = O(r^{-3-\epsilon}), \quad \epsilon > 0, \quad (326)$$

and tangent variations satisfy

$$\delta \mathcal{H}_\perp = O(r^{-3-\epsilon}), \quad \delta \mathcal{H}_i = O(r^{-3-\epsilon}). \quad (327)$$

For bounded asymptotic translation smearings these conditions make both the bulk generator and its field-space variation absolutely integrable.

Taking $R \rightarrow \infty$ therefore gives

$$\delta G[N, \beta] = \int_{\Sigma} d^3x \left(\mathcal{A}^{ij} \delta h_{ij} + \mathcal{B}_{ij} \delta \pi^{ij} \right), \quad (328)$$

with no residual surface term.

Proposition A.1 (Asymptotic differentiability). *Let $(h_{ij}, \pi^{ij}) \in \Gamma_\infty^{\text{MS}}$, and let (N, β^i) approach the asymptotic translation $(a^\perp, a^i)$ according to Eqs. (111) and (112). Then*

$$G[N, \beta] = \int_{\Sigma} d^3x \left(N \mathcal{H}_\perp + \beta^i \mathcal{H}_i \right) + a^\perp E_\infty^{\text{MS}} + a^i J_i^{\text{MS}} \quad (329)$$

is finite and functionally differentiable on the asymptotic phase space adopted in this paper.

The Regge–Teitelboim parity conditions and the elementary finiteness of the translation surface integrals play different roles. The latter follows already from

$$\bar{D}q = O(r^{-2}), \quad \pi = O(r^{-2}),$$

whereas parity removes the potentially logarithmic leading contribution to the canonical symplectic structure. The boundary completion itself does not alter the local constraint densities or their bulk Hamiltonian variations. It supplies precisely the surface primitive required when the smearing fields cease to have compact support and approach nonzero translations at infinity.

B Asymptotic Translation Estimates and Conventions

This appendix collects two technical points used repeatedly in the main text. First, we give the asymptotic estimates that imply invariance of the translation boundary coefficients under another asymptotic translation. Second, we summarize the density, sign, vector–covector, and unit conventions needed to compare the canonical boundary charge with physical energy–momentum.

The first part deliberately avoids any assertion about a global off-shell central extension of the full hypersurface-deformation algebra. Because the hypersurface-deformation bracket contains phase-space-dependent structure functions, the reduced translation statement required in Section 5 is established directly from the asymptotic Hamiltonian action on the boundary data.

Except in Subsection B.2, we use $c = 1$.

B.1 Translation action on the asymptotic data

Let

$$b = (b^\perp, b^i) \in \mathfrak{t}_\infty, \quad (330)$$

and choose an admissible representative

$$\chi_b = (N_b, \beta_b^i) \quad (331)$$

that agrees outside a compact set with the constant deformation

$$N_b = b^\perp, \quad \beta_b^i = b^i. \quad (332)$$

The asymptotic canonical data satisfy

$$h_{ij} = \delta_{ij} + O_2(r^{-1}), \quad \pi^{ij} = O_1(r^{-2}). \quad (333)$$

Hamilton's equation for the spatial metric is

$$\delta_b h_{ij} = \frac{4\kappa N_b}{\sqrt{h}} \left(\pi_{ij} - \frac{1}{2} \pi h_{ij} \right) + \mathcal{L}_{\beta_b} h_{ij}. \quad (334)$$

The momentum contribution is $O(r^{-2})$. Since the leading coordinate components of β_b^i are constant,

$$\mathcal{L}_{\beta_b} h_{ij} = b^k \partial_k h_{ij} = O(r^{-2}) \quad (335)$$

outside the compact transition region. Therefore

$$\delta_b h_{ij} = O(r^{-2}). \quad (336)$$

In particular,

$$\bar{D}_k \delta_b h_{ij} = O(r^{-3}). \quad (337)$$

For the canonical momentum, the Hamilton equation contains four types of terms:

$$\delta_b \pi^{ij} = N_b \mathcal{R}^{ij}[h] + \mathcal{D}^{ij}[N_b; h] + \mathcal{Q}^{ij}[N_b; h, \pi] + \mathcal{L}_{\beta_b} \pi^{ij}, \quad (338)$$

where $\mathcal{R}^{ij}[h]$ is proportional to $\sqrt{h} {}^{(3)}G^{ij}/\kappa$, $\mathcal{D}^{ij}$ contains the second spatial derivatives of the lapse, and $\mathcal{Q}^{ij}$ is quadratic in the canonical momentum.

The falloff

$$h_{ij} = \delta_{ij} + O_2(r^{-1})$$

implies

$${}^{(3)}G^{ij} = O(r^{-3}). \quad (339)$$

Because N_b is exactly constant sufficiently far out,

$$D_i D_j N_b = 0 \quad (340)$$

there. Moreover,

$$\mathcal{Q}^{ij} = O(r^{-4}), \quad (341)$$

while, for the weight-one density π^{ij} , the Lie derivative along the constant asymptotic shift reduces at leading order to differentiation of its asymptotic coefficients:

$$\mathcal{L}_{\beta_b}\pi^{ij} = O(r^{-3}). \quad (342)$$

Thus

$$\delta_b\pi^{ij} = O(r^{-3}). \quad (343)$$

These two estimates are stronger by one radial order than the leading canonical data. They imply that the leading r^{-1} metric coefficient and the leading r^{-2} canonical-momentum coefficient are invariant under a constant asymptotic translation, up to the proper interior action of the chosen representative.

For the temporal boundary coefficient,

$$E_\infty^{\text{MS}} = \frac{1}{2K} \lim_{R \rightarrow \infty} \oint_{S_R} (\bar{D}^j q_{ij} - \bar{D}_i q^j_j) \bar{s}^i dS_{\bar{h}}, \quad (344)$$

one therefore obtains

$$\begin{aligned} \delta_b E_\infty^{\text{MS}} &= \frac{1}{2K} \lim_{R \rightarrow \infty} \oint_{S_R} (\bar{D}^j \delta_b h_{ij} - \bar{D}_i \delta_b h^j_j) \bar{s}^i dS_{\bar{h}} \\ &= \lim_{R \rightarrow \infty} O(R^2) O(R^{-3}) = 0. \end{aligned} \quad (345)$$

Similarly,

$$J_i^{\text{MS}} = 2 \lim_{R \rightarrow \infty} \oint_{S_R} \pi_i^j dS_j \quad (346)$$

satisfies

$$\delta_b J_i^{\text{MS}} = 0. \quad (347)$$

Indeed, the contribution from $h_{ik}\delta_b\pi^{kj}$ is $O(r^{-3})$, while the extra term arising from variation of the metric used to lower the first index is

$$(\delta_b h_{ik})\pi^{kj} = O(r^{-2})O(r^{-2}) = O(r^{-4}). \quad (348)$$

Both vanish after integration over S_R in the limit.

Hence, for every $a, b \in \mathfrak{t}_\infty$,

$$\delta_b B_\infty[a] = a^\perp \delta_b E_\infty^{\text{MS}} + a^i \delta_b J_i^{\text{MS}} = 0. \quad (349)$$

The surface-deformation bracket of representatives that are constant outside a compact set has no constant asymptotic component,

$$[\chi_a, \chi_b]_{\text{SD}} = \chi_{ab}^{\text{proper}}. \quad (350)$$

Equations (349) and (350) are the only ingredients from the asymptotic generator algebra needed in the present paper. After quotienting proper deformations they yield

$$\{\mathbf{P}_\infty^{\text{MS}}(a), \mathbf{P}_\infty^{\text{MS}}(b)\}_{\text{red}} = 0. \quad (351)$$

No stronger off-shell statement about the full hypersurface-deformation algebra is required.

B.2 Density, sign, and physical-unit conventions

The spacetime and asymptotic translation-space signatures used from Paper II onward are

$$\text{sign}(g) = \text{sign}(\eta_\infty) = (-, +, +, +). \quad (352)$$

A future unit timelike vector therefore satisfies

$$\eta_\infty(u, u) = -1. \quad (353)$$

In the canonical calculation,

$$\text{wt}(h_{ij}) = 0, \quad \text{wt}(\pi^{ij}) = +1. \quad (354)$$

The canonical bracket is

$$\{h_{ij}(x), \pi^{kl}(y)\} = \delta^k_{(i} \delta^l_{j)} \delta^{(3)}(x - y). \quad (355)$$

For a contravariant vector density V^i of weight $+1$, the invariant coordinate flux is

$$\oint_S V^i dS_i, \quad dS_i := \frac{1}{2} \epsilon_{ijk} dx^j \wedge dx^k. \quad (356)$$

If s_i and dS_h are the geometric unit normal and induced area element, then

$$dS_i = \frac{s_i}{\sqrt{h}} dS_h. \quad (357)$$

This relation is essential because the canonical momentum itself already contains a factor of $\sqrt{h}$.

With the extrinsic-curvature convention used throughout the MS classical branch,

$$K_{ij} = \frac{1}{2N} (\dot{h}_{ij} - \mathcal{L}_\beta h_{ij}), \quad (358)$$

the canonical momentum is

$$\pi^{ij} = \frac{\sqrt{h}}{2\kappa} (K^{ij} - K h^{ij}). \quad (359)$$

Consequently,

$$\begin{aligned} J_i^{\text{MS}} &= 2 \lim_{R \rightarrow \infty} \oint_{S_R} \pi_i^j dS_j \\ &= \frac{1}{\kappa} \lim_{R \rightarrow \infty} \oint_{S_R} (K^j_i - K \delta^j_i) s_j dS_h. \end{aligned} \quad (360)$$

There is no additional factor of $\sqrt{h}$ in the second line.

A common ADM convention uses instead

$$K_{ij}^{\text{ADM}} = -K_{ij}. \quad (361)$$

The sign change in the spatial surface expression throughout the main text is entirely accounted for by this convention.

We now restore physical units. The canonical coefficient J_i^{MS} is energy valued and is related to the physical spatial momentum by

$$J_i^{\text{MS}} = -cP_{\infty i}^{\text{MS}}. \quad (362)$$

The energy-valued Hamiltonian translation covector is therefore

$$\mathbf{P}_{\infty}^{\text{MS}} = E_{\infty}^{\text{MS}} e^0 - cP_{\infty i}^{\text{MS}} e^i, \quad (363)$$

and satisfies

$$\mathbf{P}_{\infty}^{\text{MS}}(a) = B_{\infty}[a]. \quad (364)$$

For direct comparison with the generator convention of Refs. [5, 4], define the momentum-valued positive-response covector

$$\widehat{\Pi}_{\infty}^{\text{MS}} := \frac{1}{c} \mathbf{P}_{\infty}^{\text{MS}} = \frac{E_{\infty}^{\text{MS}}}{c} e^0 - P_{\infty i}^{\text{MS}} e^i. \quad (365)$$

The canonical action covector is

$$\Pi_{\infty}^{\text{MS}} := -\widehat{\Pi}_{\infty}^{\text{MS}} = -\frac{E_{\infty}^{\text{MS}}}{c} e^0 + P_{\infty i}^{\text{MS}} e^i. \quad (366)$$

The positive-energy four-momentum vector is

$$P_{\infty}^{\text{MS}} := \eta_{\infty}^{\#}(\Pi_{\infty}^{\text{MS}}) = -\eta_{\infty}^{\#}(\widehat{\Pi}_{\infty}^{\text{MS}}), \quad (367)$$

with components

$$P_{\infty}^{\text{MS}} = \left(\frac{E_{\infty}^{\text{MS}}}{c}, \mathbf{P}_{\infty}^{\text{MS}} \right). \quad (368)$$

Thus the convention chain is exactly the same as for a local relativistic generator:

$$P^a = \left(\frac{E}{c}, \mathbf{p} \right), \quad \Pi_a = \left(-\frac{E}{c}, \mathbf{p} \right), \quad \widehat{\Pi}_a = -\Pi_a. \quad (369)$$

The three asymptotic covector/vector objects should therefore be kept distinct:

object	components	role
$\mathbf{P}_{\infty}^{\text{MS}}$	$(E_{\infty}, -c\mathbf{P}_{\infty})$	energy-valued boundary covector
$\widehat{\Pi}_{\infty}^{\text{MS}}$	$(E_{\infty}/c, -\mathbf{P}_{\infty})$	momentum-valued positive-response covector
Π_{∞}^{MS}	$(-E_{\infty}/c, \mathbf{P}_{\infty})$	canonical action covector
P_{∞}^{MS}	$(E_{\infty}/c, \mathbf{P}_{\infty})$	positive-energy four-vector

(370)

In particular,

$$\widehat{\Pi}_{\infty}^{\text{MS}} = \frac{1}{c} \mathbf{P}_{\infty}^{\text{MS}} = -\Pi_{\infty}^{\text{MS}}, \quad P_{\infty}^{\text{MS}} = -\eta_{\infty}^{\#}(\widehat{\Pi}_{\infty}^{\text{MS}}). \quad (371)$$

The relation

$$P_{\infty}^{\text{MS}} = -\eta_{\infty}^{\#}(\mathbf{P}_{\infty}^{\text{MS}})$$

is therefore valid only after $c = 1$ has been imposed; it should not be used in dimensionful formulas.

With

$$\kappa = \frac{8\pi G}{c^4}, \quad (372)$$

the physical surface coefficients are

$$E_{\infty}^{\text{MS}} = \frac{c^4}{16\pi G} \lim_{R \rightarrow \infty} \oint_{S_R} (\bar{D}^j q_{ij} - \bar{D}_i q^j{}_j) \bar{s}^i dS_{\bar{h}}, \quad (373)$$

and

$$P_{\infty i}^{\text{MS}} = -\frac{c^3}{8\pi G} \lim_{R \rightarrow \infty} \oint_{S_R} (K^j{}_i - K \delta^j{}_i) s_j dS_h. \quad (374)$$

Using $K_{ij}^{\text{ADM}} = -K_{ij}$, the second expression takes the conventional ADM sign.

Finally, if P_{∞}^{MS} is future timelike, its momentum-valued Lorentzian radius is

$$Q_{\infty}^{\text{MS}} = \sqrt{-\eta_{\infty}(P_{\infty}^{\text{MS}}, P_{\infty}^{\text{MS}})} = \sqrt{\left(\frac{E_{\infty}^{\text{MS}}}{c}\right)^2 - |\mathbf{P}_{\infty}^{\text{MS}}|^2}, \quad (375)$$

and

$$M_{\infty}^{\text{MS}} = \frac{Q_{\infty}^{\text{MS}}}{c}. \quad (376)$$

The dimensions are therefore

$$[E_{\infty}] = \text{energy}, \quad [P_{\infty i}] = [Q_{\infty}] = \text{momentum}, \quad [M_{\infty}] = \text{mass}, \quad (377)$$

while D and S remain dimensionless.

C Observer-Space Reconstruction and Calibration Quotients

This appendix records two supplementary structures that are useful for interpreting the main results but are not needed for the boundary construction itself. The first gives an intrinsic characterization of the observer-energy functions generated by ambient Lorentzian translation vectors. The second makes precise the distinction between strong response calibration and the weaker condition of preserving only the four translation charges.

Both structures are standard at the level of their underlying geometry or linear algebra. They are included because they clarify how the observer and response descriptions used in the main text compress the four translation degrees of freedom.

C.1 Observer-space reconstruction from the Hessian equation

Let

$$(V, \eta) \simeq \mathbb{R}^{1,3}, \quad \text{sign}(\eta) = (-, +, +, +), \quad (378)$$

and let

$$\mathcal{H}^+ := \{u \in V : \eta(u, u) = -1, \quad u \text{ future directed}\} \quad (379)$$

carry the induced hyperbolic metric h_H .

Every ambient Lorentz vector $P \in V$ defines the scalar function

$$E_P(u) := -\eta(P, u). \quad (380)$$

As used in Section 6,

$$\nabla_H E_P = -P + E_P u, \quad (381)$$

and hence

$$P = E_P u - \nabla_H E_P. \quad (382)$$

The corresponding Hessian equation is

$$\text{Hess}_H E_P = E_P h_H. \quad (383)$$

The converse is equally useful.

Proposition C.1 (Observer-space characterization of ambient translations). *Let $f \in C^\infty(\mathcal{H}^+)$ satisfy*

$$\text{Hess}_H f = f h_H. \quad (384)$$

Then there exists a unique $P_f \in V$ such that

$$f(u) = -\eta(P_f, u) \quad (385)$$

for all $u \in \mathcal{H}^+$. At every observer,

$$P_f = f u - \nabla_H f. \quad (386)$$

Consequently, the solution space of Eq. (384) is four dimensional and is naturally isomorphic to the ambient translation space V .

Proof. Regard $\mathcal{H}^+$ as the unit hyperboloid embedded in V , and let D denote the ambient flat connection. For tangent vector fields X, Y on $\mathcal{H}^+$, the Gauss formula is

$$D_X Y = \nabla_X^H Y + h_H(X, Y)u. \quad (387)$$

Define the ambient vector field along the hyperboloid

$$\mathcal{P}_f := f u - \nabla_H f. \quad (388)$$

For any tangent X ,

$$\begin{aligned} D_X \mathcal{P}_f &= X(f)u + fX - D_X \nabla_H f \\ &= X(f)u + fX - \nabla_X^H \nabla_H f - h_H(X, \nabla_H f)u. \end{aligned} \quad (389)$$

Equation (384) gives

$$\nabla_X^H \nabla_H f = fX,$$

while

$$h_H(X, \nabla_H f) = X(f).$$

Therefore

$$D_X \mathcal{P}_f = 0. \quad (390)$$

The vector field $\mathcal{P}_f$ is consequently the restriction of a single constant ambient vector $P_f \in V$.

Since $\eta(u, u) = -1$ and $\nabla_H f$ is tangent to the hyperboloid,

$$\eta(P_f, u) = -f, \quad (391)$$

which proves Eq. (385).

Uniqueness follows because a Lorentz vector orthogonal to every future unit timelike vector must vanish. The map

$$P \mapsto -\eta(P, \cdot)|_{\mathcal{H}^+}$$

is therefore a linear bijection from the four-dimensional space V to the solution space of Eq. (384). $\square$

This proposition is the standard static-potential characterization of hyperbolic space. It is not claimed as a new mathematical result. In the present setting its role is to sharpen the observer-space interpretation: once the boundary construction has supplied a total translation charge, the corresponding scalar observer-energy function may be characterized without reference to a particular inertial basis by

$$\text{Hess}_H \mathcal{E}_\infty = \mathcal{E}_\infty h_H. \quad (392)$$

The complete asymptotic translation vector is then reconstructed from the first observer-space jet,

$$P_\infty^{\text{MS}} = \mathcal{E}_\infty u - \nabla_H \mathcal{E}_\infty. \quad (393)$$

The Lorentzian norm is likewise determined at any observer:

$$-\eta(P_f, P_f) = f^2 - |\nabla_H f|_{h_H}^2. \quad (394)$$

Equation (384) makes the right-hand side constant, since

$$\begin{aligned} d(f^2 - |\nabla_H f|_{h_H}^2) &= 2f df - 2 \text{Hess}_H f (\nabla_H f, \cdot) \\ &= 0. \end{aligned} \quad (395)$$

The causal class of the ambient vector can therefore be read directly from the sign of this observer-space first integral.

C.2 Translation-visible response jets and reference stability

Section 7 showed that the leading translation-sensitive response recalibration is described by

$$\lambda - 1 = \frac{\sigma(\hat{x})}{r} + o(r^{-1}), \quad \sigma(-\hat{x}) = \sigma(\hat{x}), \quad (396)$$

together with the normal first jet

$$n(\ln \lambda) = \frac{\rho(\hat{x})}{r^2} + o(r^{-2}), \quad \rho(-\hat{x}) = -\rho(\hat{x}). \quad (397)$$

At these orders the induced translation-charge shifts are

$$\Delta_\lambda E_\infty^{\text{MS}} = \frac{2}{\kappa} \int_{S^2} \sigma d\Omega, \quad (398)$$

and

$$\Delta_\lambda J_i^{\text{MS}} = -\frac{2}{\kappa} \int_{S^2} \rho \hat{x}_i d\Omega. \quad (399)$$

It is useful to formulate these relations as a quotient of the leading response-jet space. Define

$$\mathcal{J}_{\text{tr}} := C_{\text{even}}^\infty(S^2) \oplus C_{\text{odd}}^\infty(S^2), \quad (400)$$

whose elements are pairs (σ, ρ) , and introduce the linear translation-readout map

$$C_{\text{tr}} : \mathcal{J}_{\text{tr}} \longrightarrow \mathbb{R} \oplus \mathbb{R}^3 \quad (401)$$

by

$$C_{\text{tr}}(\sigma, \rho) := \left(\int_{S^2} \sigma d\Omega, \int_{S^2} \rho \hat{x}_i d\Omega \right). \quad (402)$$

Overall constants such as $2/\kappa$ have been omitted from this purely representation-theoretic map.

The map is surjective. A constant σ produces an arbitrary scalar component, while

$$\rho(\hat{x}) = v_i \hat{x}^i \quad (403)$$

gives

$$\int_{S^2} \rho(\hat{x}) \hat{x}_i d\Omega = \frac{4\pi}{3} v_i. \quad (404)$$

Hence the first isomorphism theorem gives

$$\mathcal{J}_{\text{tr}} / \ker C_{\text{tr}} \simeq \mathbb{R} \oplus \mathbb{R}^3. \quad (405)$$

Under the rotation group this quotient is

$$\mathcal{J}_{\text{tr}} / \ker C_{\text{tr}} \simeq \mathbf{1} \oplus \mathbf{3}. \quad (406)$$

In spherical harmonics,

$$\sigma = \sigma_{l=0} + \sigma_{l \geq 2, \text{ even}}, \quad (407)$$

whereas

$$\rho = \rho_{l=1} + \rho_{l \geq 3, \text{ odd}}. \quad (408)$$

The kernel of the translation readout contains precisely the modes that have vanishing $l = 0$ projection in σ and vanishing $l = 1$ projection in ρ . The translation sector therefore retains only

$$l = 0 \oplus l = 1 \quad (409)$$

from the displayed response jet.

Equation (405) is not a new representation-theoretic result. It is a concise formulation of the operational content of Eqs. (398) and (399): within the MS response formulation, the four translation charges are exactly the four-dimensional quotient of the leading calibration data visible to asymptotic translations.

This distinction also clarifies the strong and weak calibration conditions. The strong first-jet calibration used in the definition of the main phase space,

$$\lambda = 1 + o(r^{-1}), \quad \bar{D}_i \lambda = o(r^{-2}), \quad n(\ln \lambda) = o(r^{-2}), \quad (410)$$

sets the entire leading element of $\mathcal{J}_{\text{tr}}$ to zero. It therefore removes all residual response-scale freedom at the translation-supporting asymptotic orders.

If only the four integrated translation charges are required to remain unchanged, it is sufficient at the displayed leading orders to impose

$$(\sigma, \rho) \in \ker C_{\text{tr}}, \quad (411)$$

or explicitly

$$\int_{S^2} \sigma d\Omega = 0, \quad \int_{S^2} \rho \hat{x}_i d\Omega = 0. \quad (412)$$

The strong condition is therefore a pointwise removal of all charge-order calibration freedom, whereas the weak condition removes only its projection onto the four-dimensional translation quotient.

Finally, the response calibration should be distinguished from small changes of the calibrated reference that occur strictly below the charge-supporting metric order. Let $\bar{h}_{ij}$ and $\bar{h}'_{ij}$ be two calibrated flat reference metrics satisfying

$$\bar{h}'_{ij} - \bar{h}_{ij} = o(r^{-1}), \quad \bar{D}_k(\bar{h}'_{ij} - \bar{h}_{ij}) = o(r^{-2}). \quad (413)$$

The change in the temporal boundary functional is

$$\Delta_{\text{ref}} E_{\infty}^{\text{MS}} = -\frac{1}{2\kappa} \lim_{R \rightarrow \infty} \oint_{S_R} \bar{s}^i \left[\bar{D}^j (\bar{h}'_{ij} - \bar{h}_{ij}) - \bar{D}_i (\bar{h}'^j_j - \bar{h}^j_j) \right] dS_{\bar{h}}. \quad (414)$$

The integrand is $o(r^{-2})$, so

$$\Delta_{\text{ref}} E_{\infty}^{\text{MS}} = 0. \quad (415)$$

Thus reference changes strictly below the r^{-1} metric order are invisible to the energy charge.

The spatial translation coefficient in the phase space adopted here is defined directly from the physical K_{ij} or π^{ij} , with the calibrated reference slicing fixed by

$$\bar{K}_{ij} = 0.$$

Consequently, there is no independent reference- K subtraction in the formula for J_i^{MS} . A change that modifies the leading physical normal-boost data at order r^{-2} , however, is not below charge-supporting order and can change the spatial translation charge, exactly as described by Eq. (399).

The asymptotic hierarchy can therefore be summarized as

changes below charge-supporting order $\longrightarrow$ no translation-charge change,

$$\ker C_{\text{tr}} \longrightarrow \text{translation-invisible leading response jets}, \quad (416)$$

$$\mathcal{J}_{\text{tr}} / \ker C_{\text{tr}} \longrightarrow \text{four charge-supporting response modes}.$$

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